MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

GATE, EXCLUSIVE OR, DIGITAL, KEY CODE BMC

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic exclusive OR gate module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.
MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.0776 failures per million operating hours for class I or 0.3405 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be BMC.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/255 -1 for class I and M28787/255 -2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 3 to 18V ±5%.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

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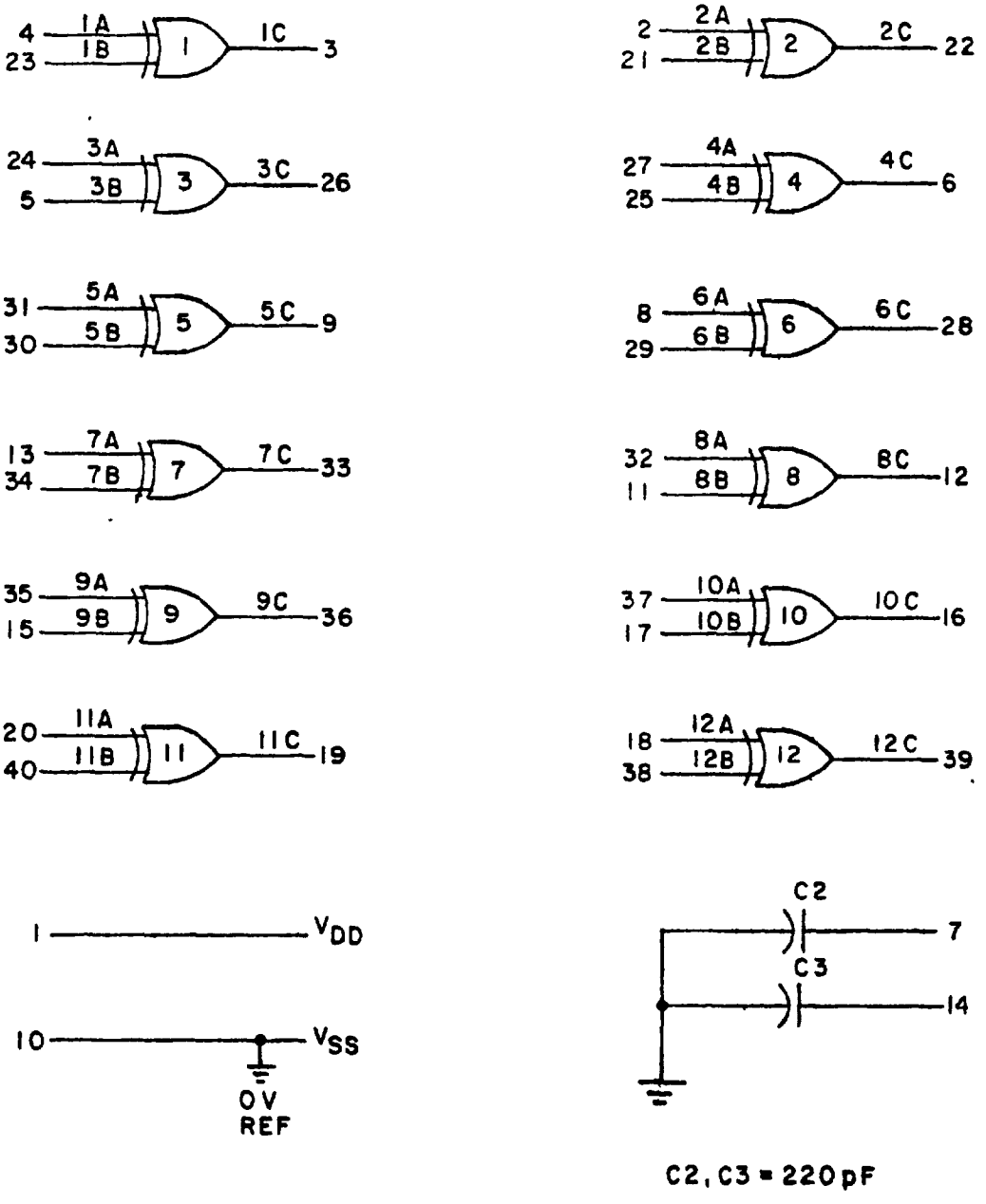


FIGURE 1. BMC module function diagram, exclusive OR gate.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation Initial End-of-life	III III	S S	10 1	∞ ∞	10 1	∞ ∞	10 1	∞ ∞	10 1	∞ ∞	10 1	∞ ∞	MΩ MΩ
B1-B24	Input clamp diode voltage, negative	IV	X	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	V
C1-C24	Input clamp diode voltage, positive	V	X	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	V
D1-D12	Static output voltage, low level (I_{OL}= 0)	VI	X	0.00	0.05	0.00	0.05	0.00	0.05	0.00	0.05	0.00	0.05	V
E1-E12	Static output voltage, high level (I_{OH} = 0)	VII	X	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	V
F1-F24	Output voltage, low level, V_{DD} = 5V	VIII	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
G1-G24	Output voltage, high level, V_{DD} = 5V	IX	X	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	V
H1-H12	Output voltage, low level, V_{DD} = 15V	X	X	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	V
J1-J12	Output voltage, high level, V_{DD} = 15V	XI	X	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	V
K1-K24	Input current, low level	XII	X	0	-200	---	---	0	-200	---	---	0	-200	nA
L1-L24	Input current, high level	XIII	X	0	200	---	---	0	200	---	---	0	200	nA
M1	Power supply current, low level	XIV	X	0	6	0	6	0	60	0	6	0	120	µA
M2	Power supply current, high level	XIV	X	0	6	0	6	0	60	0	6	0	120	µA
N1-N12	Transition time (t_{TLH}), C_L= 50 pF, V_{DD}= 5V	XV	S	20	200	20	200	22	220	15	150	24	245	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
N13-N24	Transition time (t_{TLH}), C_L = 150 pF, V_{DD} = 5V	XV	Q	68	680	68	680	75	750	51	515	83	835	ns
P1-P12	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 5V	XVI	S	20	200	20	200	22	220	15	150	24	245	ns
P13-P24	Transition time (t_{THL}), C_L = 150 pF, V_{DD} = 5V	XVI	Q	68	680	68	680	75	750	51	515	83	835	ns
Q1-Q12	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 10V	XVII	Q	10	100	10	100	11	110	7	75	12	125	ns
R1-R12	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 10V	XVIII	Q	10	100	10	100	11	110	7	75	12	125	ns
S1-S12	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 15V	XIX	Q	8	80	8	80	9	90	6	60	10	100	ns
T1-T12	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 15V	XX	Q	8	80	8	80	9	90	6	60	10	100	ns
U1-U24	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5V	XXI	S	35	350	35	350	38	385	26	265	43	430	ns
U25-U48	Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5V	XXI	Q	55	550	55	550	61	610	42	420	67	675	ns
V1-V24	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5V	XXII	S	35	350	35	350	38	385	26	265	43	430	ns
V25-V48	Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5V	XXII	Q	55	550	55	550	61	610	42	420	67	675	ns
W1-W24	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10V	XXIII	Q	15	150	15	150	16	165	11	115	18	185	ns
X1-X24	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10V	XXIV	Q	15	150	15	150	16	165	11	115	18	185	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
Y1-Y24	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15V	XXV	Q	10	100	10	100	11	110	7	75	12	125	ns
Z1-Z24	Propagation delay time, (t_{PHL}), C_L = 50 pF, V_{DD} = 15V	XXVI	Q	10	100	10	100	11	110	7	75	12	125	ns
AA	Power supply filter capacitance	XXVII	S	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	µF
BB1, BB2	C2 and C3 capacitance	XXVIII	X	176	264	176	264	176	264	176	264	176	264	pF
CC1-CC24	Input capacitance	XXIX	Q	2	15	2	15	2	15	2	15	2	15	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent, or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through CC shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through CC in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B through J and M through CC shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2. Test conditions K and L shall be performed at 25°, 60° and 100°C only using the initial limits for the applicable operating temperature.

4.5.1.1 Transient thermal test. For the transient thermal tests, use the same test conditions as the maximum class temperature conditions.

4.5.1.2 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition AA, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests B through BB shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests B through CC shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements: The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

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6.4 Module description. The following electrical parameters reflect the 0° to 60°C and -55° to 100°C end-of-life limits from table I.

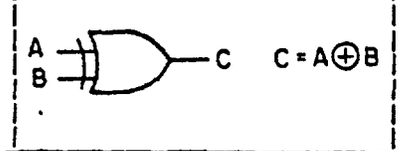
	Class I	Class II	Unit
Maximum power dissipation	1.8	1.8	mW
Input voltage			
Low level (V_{DD} = 5V)	0.0 min, 1.5 max	0.0 min, 1.5 max	V
(V_{DD} = 15V)	0 min, 4 max	0 min, 4 max	V
High level (V_{DD} = 5V)	3.5 min, V_{DD} max	3.5 min, V_{DD} max	V
(V_{DD} = 15V)	11 min, V_{DD} max	11 min, V_{DD} max	V
Input current			
Low level (V_{IL} = 0V, V_{DD} = 18V)	0 min, -200 max	0 min, -200 max	nA
High level (V_{IH} = 18V, V_{DD} = 18V)	0 min, 200 max	0 min, 200 max	nA
Output drive current (V_{DD} = 5V)			
Low level (V_{OL} = 0.4V)			
I_{OL} = at -55°C	---	0.64 min	mA
0° and 25°C	0.51 min	0.51 min	mA
60°C	0.44 min	0.44 min	mA
100°C	---	0.36 min	mA
High level (V_{OH} = 4.6V)			
I_{OH} = at -55°C	---	-0.64 min	mA
0° and 25°C	-0.51 min	-0.51 min	mA
60°C	-0.44 min	-0.44 min	mA
100°C	---	-0.36 min	mA

6.5 Module function. The module contains 12 independent exclusive OR gates. This module can be used as even and odd parity generators, checkers and comparators. There are two 220 pF capacitors, with one side connected to ground, for external use.

6.5.1 Input to output relationship. The outputs shall have the input to output relationship of table II for positive logic.

TABLE II. Input to output relationship for positive logic. 1/

Inputs		Output
A	B	C
L	L	L
H	L	H
L	H	H
H	H	L



1/ H = high level.
L = low level.

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6.5.2 Paralleling of outputs. The paralleling of outputs to obtain extended drive and sink capability is permissible only when parallel inputs and outputs are confined within each group specified below. The output drive capability of this combination is equal to the sum of the output drive capabilities of the individual circuits. The circuit groups which can be paralleled are:

a. 1, 2, 3, 4.

b. 5, 6, 7, 8.

c. 9, 10, 11, 12.

6.5.3 Dynamic power dissipation. The dynamic power dissipation can be approximated by the formula:

$$P_{TOTAL} = (C\ V_{DD}{}^{2}f)N + \text{Quiescent power dissipation.}$$

Where C = Load capacitance.
V_{DD} = Supply voltage.
f = operating frequency.
N = Number of outputs.

6.6 Module handling considerations. To avoid possible damage to the devices during handling, testing, or actual operation, the following procedures should be followed:

a. Device pins should be in contact with conductive material, except when being tested or in actual operation, to avoid build-up of static charge.

b. Soldering iron tips, metal parts of fixtures and tools, and handling facilities should be grounded.

c. Signals should not be applied to the inputs while the device power supply is off.

d. All input signals and conditioning voltages must be removed before V_{DD} is disconnected.

e. Input voltages greater than the supply (V_{DD} - V_{SS}) should not be applied as this will forward bias the protection input diodes.

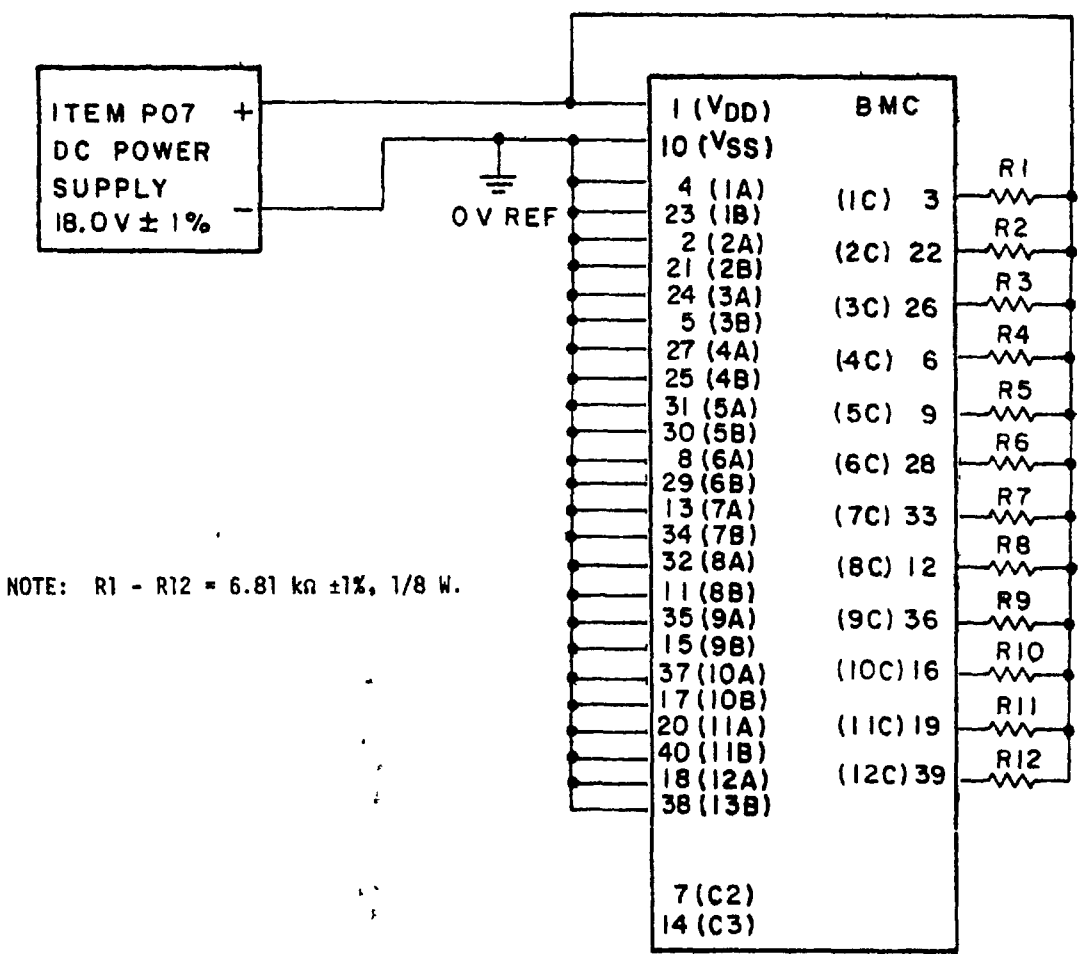
f. All unused input leads must be connected to either V_{SS} (GND) or V_{DD} (device supply), whichever is appropriate for the logic circuit involved.

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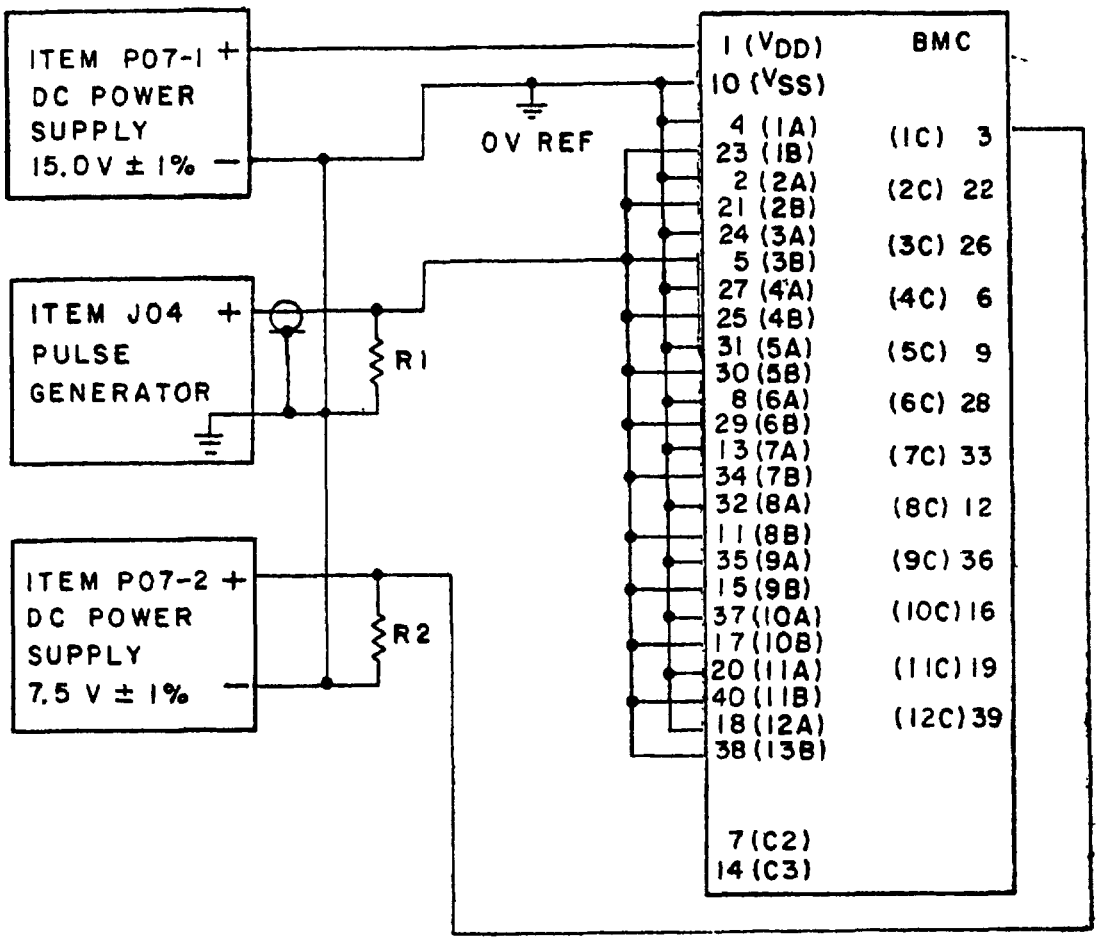
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NOTE: R1 - R12 = 6.81 kΩ ±1%, 1/8 W.

FIGURE 2. Operating temperature and life test circuit.

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NOTES:
1. R1 = R2 = 51.1Ω ±5%, 5 W.
2. Pulse generator setup:
   PW = 500 ±100 ns.
   Frequency = 1.0 MHz ±20%.
   $t_{TLH} = t_{THL} = 50 \pm 10$ ns.
   PA = 15^{+0}_{-1} V.

FIGURE 3. Component temperature test circuit.

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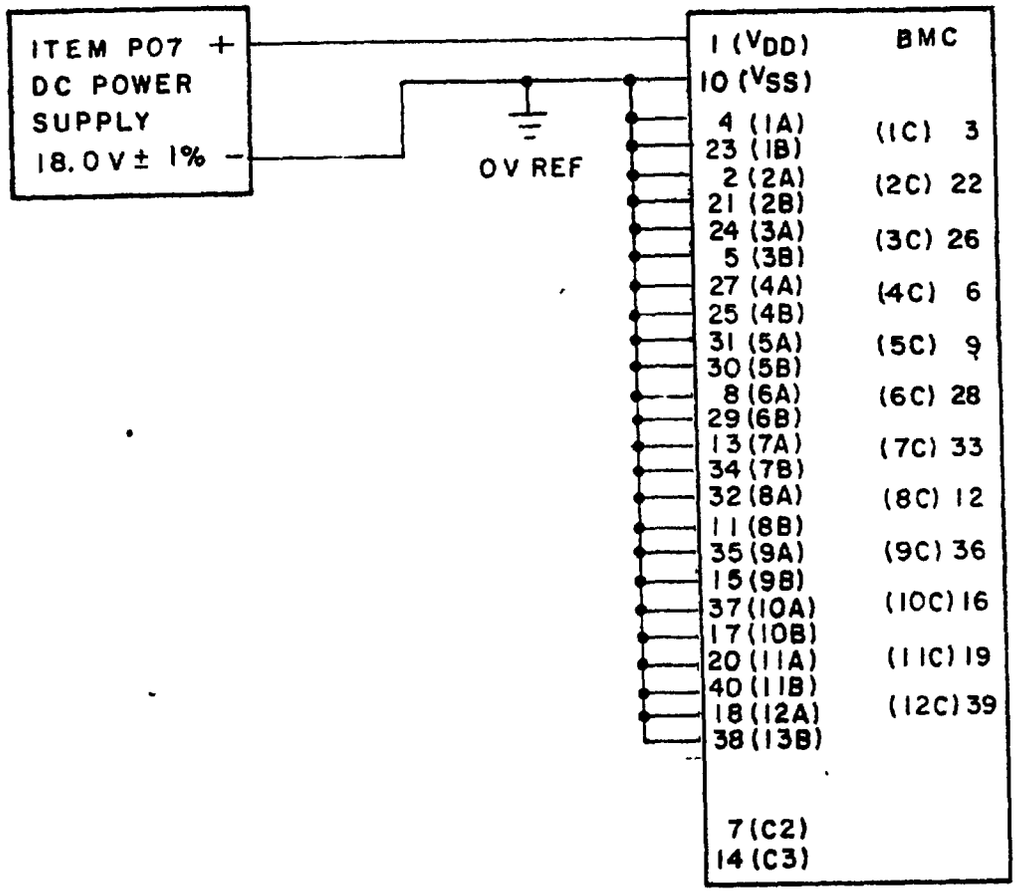


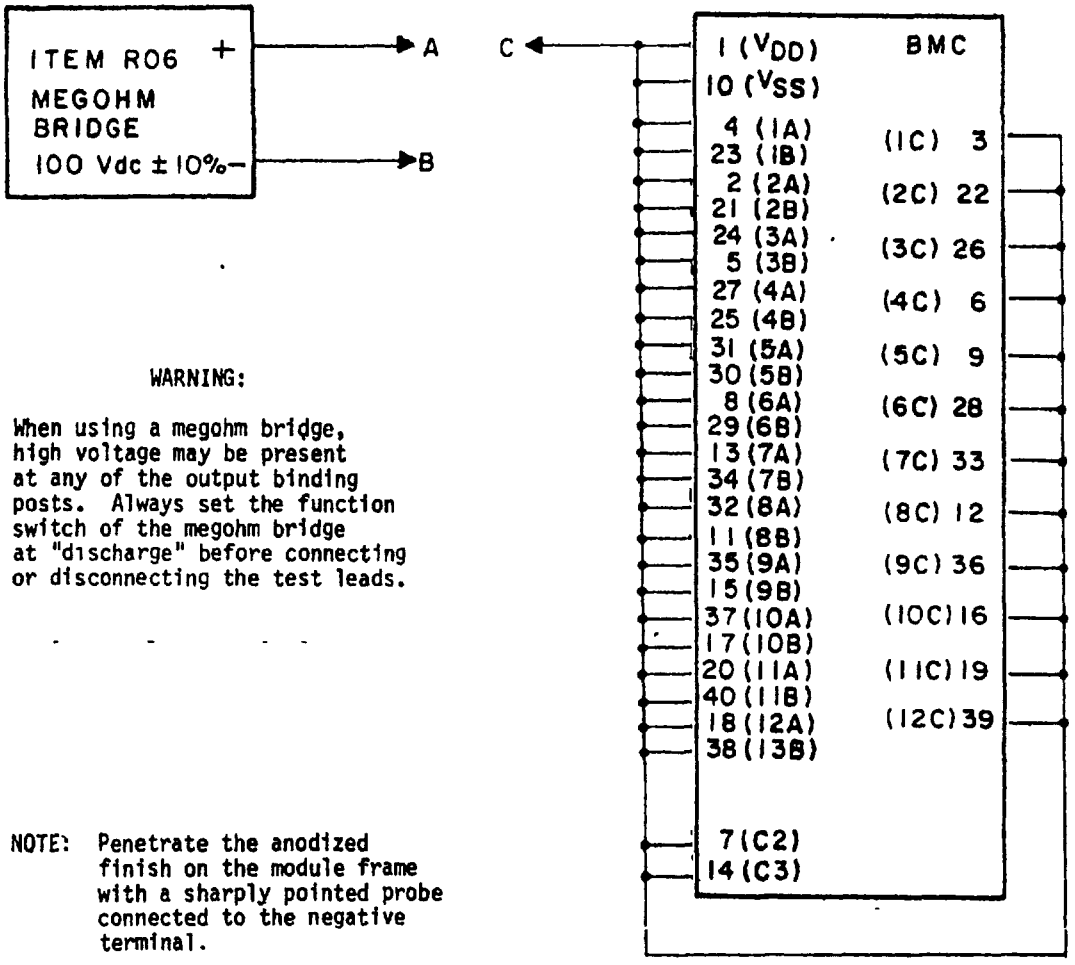
FIGURE 4. Vibration test circuit.

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TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation	C	Frame	10	∞	MΩ	5

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the function switch of the megohm bridge at "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 5. Isolation test circuit for table III.

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TABLE IV. Input clamp diode voltage (negative) test.

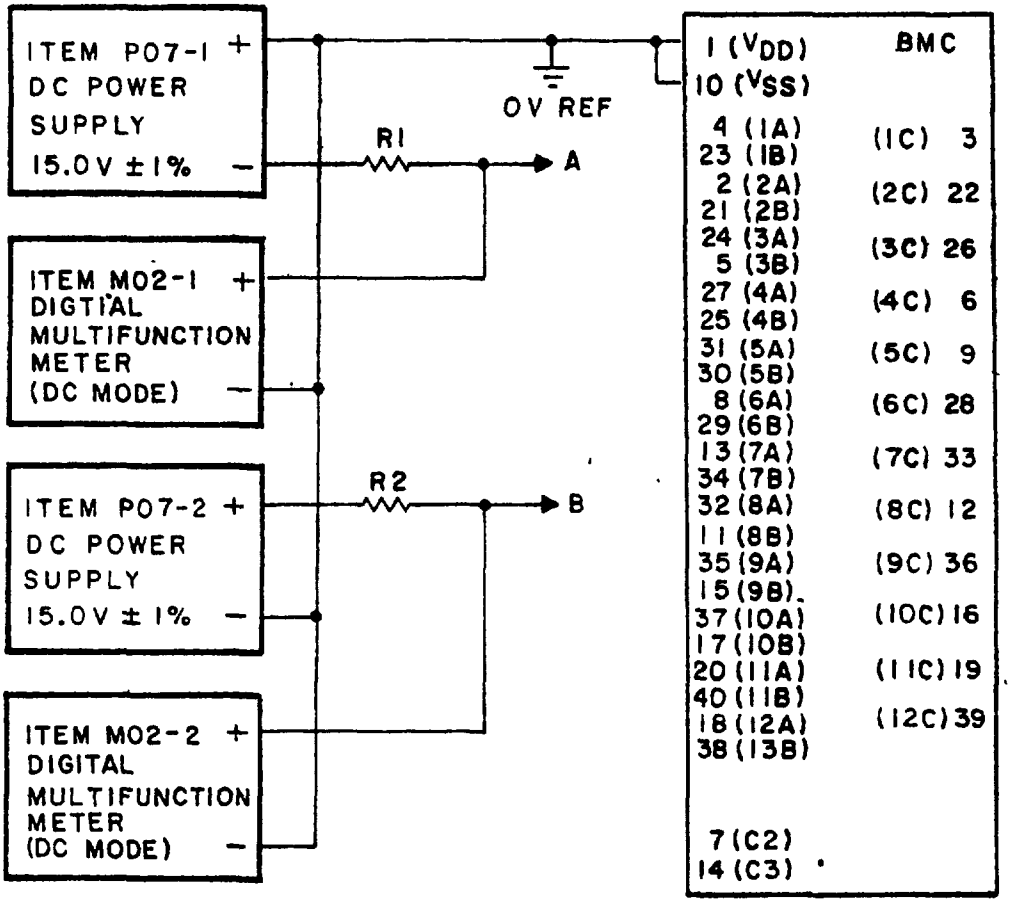
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	1A Input clamp diode voltage, negative	4	Open	-0.4	-6.0	V	6
B2	1B	23					
B3	2A	2					
B4	2B	21					
B5	3A	24					
B6	3B	5					
B7	4A	27					
B8	4B	25					
B9	5A	31					
B10	5B	30					
B11	6A	8					
B12	6B	29					
B13	7A	13					
B14	7B	34					
B15	8A	32					
B16	8B	11					
B17	9A	35					
B18	9B	15					
B19	10A	37					
B20	10B	17					
B21	11A	20					
B22	11B	40					
B23	12A	18					
B24	12B	38					

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TABLE V. Input clamp diode voltage (positive) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	1A Input clamp diode voltage, positive	Open	4	0.4	1.5	V	6
C2	1B		23				
C3	2A		2				
C4	2B		21				
C5	3A		24				
C6	3B		5				
C7	4A		27				
C8	4B		25				
C9	5A		31				
C10	5B		30				
C11	6A		8				
C12	6B		29				
C13	7A		13				
C14	7B		34				
C15	8A		32				
C16	8B		11				
C17	9A		35				
C18	9B		15				
C19	10A		37				
C20	10B		17				
C21	11A		20				
C22	11B		40				
C23	12A		18				
C24	12B		38				

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NOTES:
1. R1 = 9.09 kΩ ±1%, 1/8 W.
2. R2 = 13.7 kΩ ±1%, 1/8 W.
3 Since items P07-1, P07-2, M02-1, and M02-2 are not used simultaneously, only one item P07 and one item M02 is required.

FIGURE 6. Input clamp diode voltage (negative and positive) test circuit for tables IV and V.

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TABLE VI. Static output voltage (low-level, I_{OL} = 0) test.

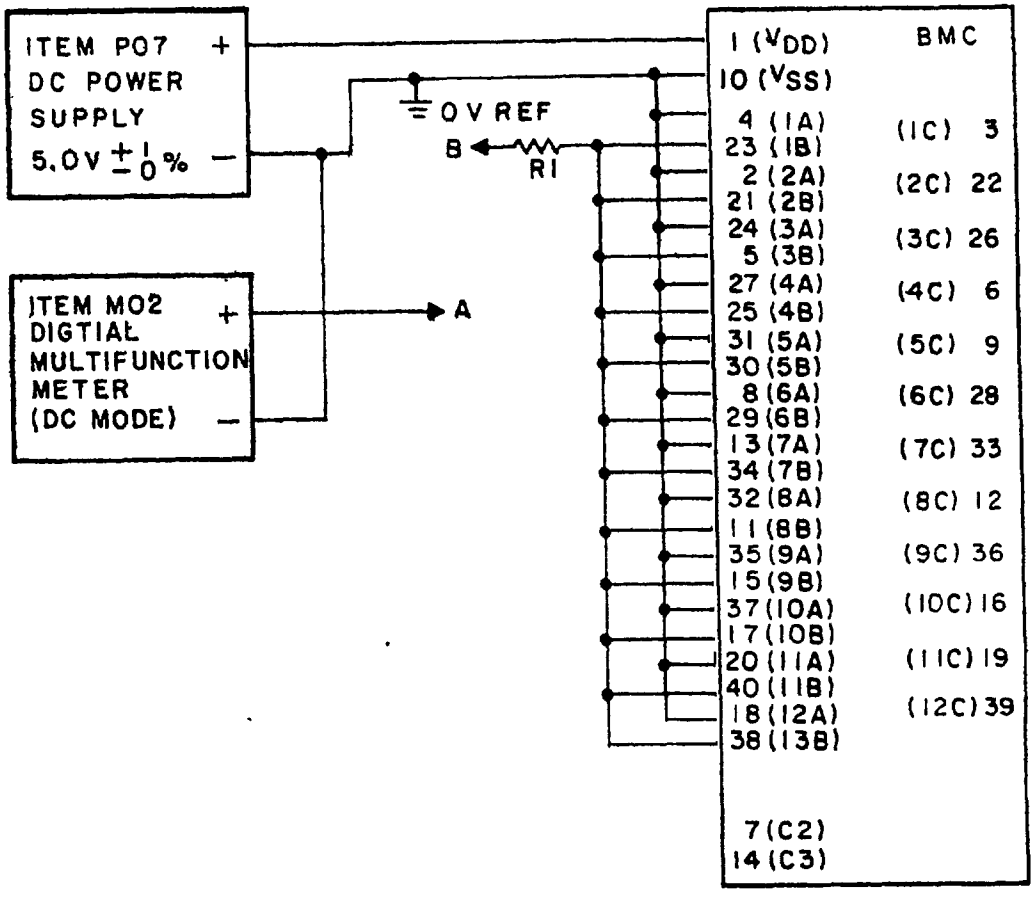
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D1	1C Static output voltage, low-level (I_{OL} = 0)	3	10	0.00	0.05	V	7
D2	2C	22					
D3	3C	26					
D4	4C	6					
D5	5C	9					
D6	6C	28					
D7	7C	33					
D8	8C	12					
D9	9C	36					
D10	10C	16					
D11	11C	19					
D12	12C	39					

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TABLE VII. Static output voltage (high-level, I_{OH} = 0) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
E1	1C Static output voltage, high-level (I_{OH} = 0)	3	1	4.95	5.05	V	7
E2	2C	22					
E3	3C	26					
E4	4C	6					
E5	5C	9					
E6	6C	28					
E7	7C	33					
E8	8C	12					
E9	9C	36					
E10	10C	16					
E11	11C	19					
E12	12C	39					

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NOTE: R1 = 1.5 kΩ ±1%, 1/4 W.

FIGURE 7. Static output voltage (low-level, I_{OL} = 0, and high-level, I_{OH} = 0) test circuit for tables VI and VII.

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TABLE VIII. Output voltage (low-level, V_{DD} = 5V) test.

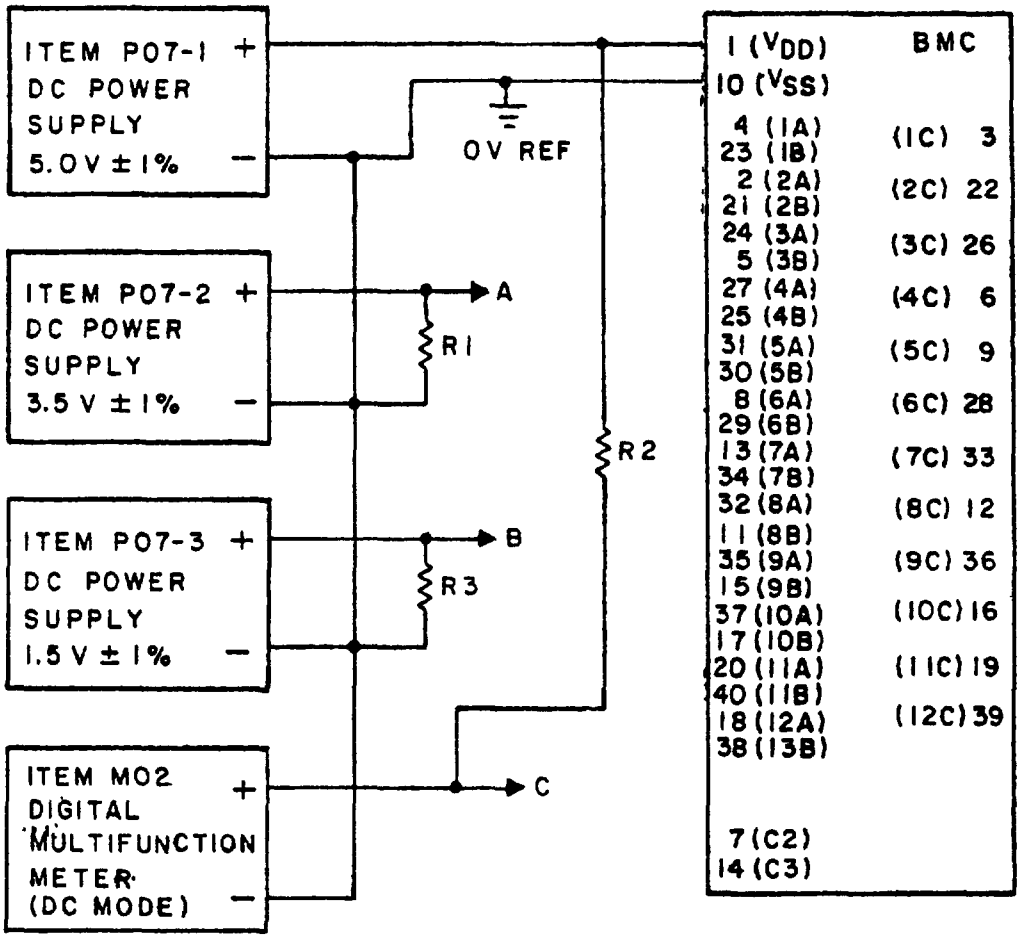
Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
F1	1C Output voltage, low level, V_{DD} = 5V	21,5,25,30,29,34,11,15,17,40,38	4,23,2,24,27,31,8,13,32,35,37,20,18	3	0.0	0.4	V	8
F2	2C	23,5,25,30,29,34,11,15,17,40,38	2,21,4,24,27,31,8,13,32,35,37,20,18	22				
F3	3C	23,21,25,30,29,34,11,15,17,40,38	24,5,4,2,27,31,8,13,32,35,37,20,18	26				
F4	4C	23,21,5,30,29,34,11,15,17,40,38	27,25,4,2,24,31,8,13,32,35,37,20,18	6				
F5	5C	23,21,5,25,29,34,11,15,17,40,38	31,30,4,2,24,27,8,13,32,35,37,20,18	9				
F6	6C	23,21,5,25,30,34,11,15,17,40,38	8,29,4,2,24,27,31,13,32,35,37,20,18	28				
F7	7C	23,21,5,25,30,29,11,15,17,40,38	13,34,4,2,24,27,31,8,32,35,37,20,18	33				
F8	8C	23,21,5,25,30,29,34,15,17,40,38	32,11,4,2,24,27,31,8,13,35,37,20,18	12				
F9	9C	23,21,5,25,30,29,34,11,17,40,38	35,15,4,2,24,27,31,8,13,32,37,20,18	36				
F10	10C	23,21,5,25,30,29,34,11,15,40,38	37,17,4,2,24,27,31,8,13,32,35,20,18	16				
F11	11C	23,21,5,25,30,29,34,11,15,17,38	20,40,4,2,24,27,31,8,13,32,35,37,18	19				
F12	12C	23,21,5,25,30,29,34,11,15,17,	18,38,4,2,24,27,31,8,13,32,35,37,20	39				

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TABLE VIII. Output voltage (low-level, V_{DD} = 5V) test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
F13	1C Output voltage, low level, V_{DD} = 5V	4,23,21,5,25,30,29,34,11,15,17,40,38	2,24,27,31,8,13,32,35,37,20,18	3	0.0	0.4	V	8
F14	2C	2,21,23,5,25,30,29,34,11,15,17,40,38	4,24,27,31,8,13,32,35,37,20,18	22				
F15	3C	24,5,23,21,25,30,29,34,11,15,17,40,38	4,2,27,31,8,13,32,35,37,20,18	26				
F16	4C	27,25,23,21,5,30,29,34,11,15,17,40,38	4,2,24,31,8,13,32,35,37,20,18	6				
F17	5C	31,30,23,21,5,25,29,34,11,15,17,40,38	4,2,24,27,8,13,32,35,37,20,18	9				
F18	6C	8,29,23,21,5,25,30,34,11,15,17,40,38	4,2,24,27,31,13,32,35,37,20,18	28				
F19	7C	13,34,23,21,5,25,30,29,11,15,17,40,38	4,2,24,27,31,8,32,35,37,20,18	33				
F20	8C	32,11,23,21,5,25,30,29,34,15,17,40,38	4,2,24,27,31,8,13,35,37,20,18	12				
F21	9C	35,15,23,21,5,25,30,29,34,11,17,40,38	4,2,24,27,31,8,13,32,37,20,18	36				
F22	10C	37,17,23,21,5,25,30,29,34,11,15,40,38	4,2,24,27,31,8,13,32,35,20,18	16				
F23	11C	20,40,23,21,5,25,30,29,34,11,15,17,38	4,2,24,27,31,8,13,32,35,37,18	19				
F24	12C	18,38,23,21,5,25,30,29,34,11,15,17,40	4,2,24,27,31,8,13,32,35,37,20	39				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:06Z
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NOTES:
1. R1 = R3 = 51.1Ω ±5%, 1/2 W.
2. R2 = @ -55°C 7.15 kΩ ±1%, 1/8 W.
   @ 0° and 25° 9.09 kΩ ±1%, 1/8 W.
   @ 60°C 10.5 kΩ ±1%, 1/8 W.
   @ 100°C 12 7 kΩ ±1%, 1/8 W.

FIGURE 8. Output voltage (low-level, V_{DD} = 5 V) test circuit for table VIII.

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TABLE IX. Output voltage (high-level, V_{DD} = 5V) test.

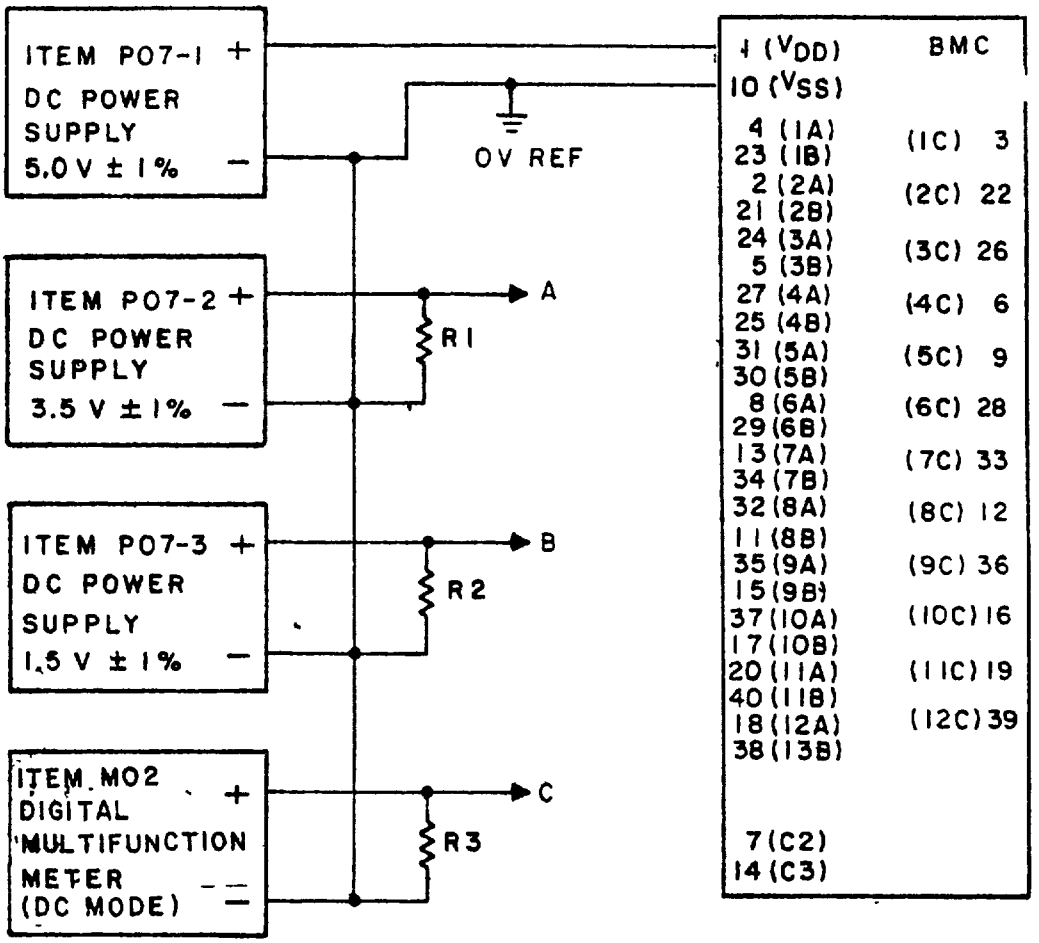
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G1	1C Output voltage, high level, V_{DD} = 5V	4	23,2,21,24,5,27,25,31,30,8,29,13,34, 32,11,35,15,37,17,20,40,18,38	3	4.6	5.0	V	9
G2	2C	2	4,23,21,24,5,27,25,31,30,8,29,13,34, 32,11,35,15,37,17,20,40,18,38	22				
G3	3C	24	4,23,2,21,5,27,25,31,30,8,29,13,34, 32,11,35,15,37,17,20,40,18,38	26				
G4	4C	27	4,23,2,21,24,5,25,31,30,8,29,13,34, 32,11,35,15,37,17,20,40,18,38	6				
G5	5C	31	4,23,2,21,24,5,27,25,30,8,29,13,34, 32,11,35,15,37,17,20,40,18,38	9				
G6	6C	8	4,23,2,21,24,5,27,25,31,30,29,13,34, 32,11,35,15,37,17,20,40,18,38	28				
G7	7C	13	4,23,2,21,24,5,27,25,31,30,8,29,34, 32,11,35,15,37,17,20,40,18,38	33				
G8	8C	32	4,23,2,21,24,5,27,25,31,30,8,29,13, 34,11,35,15,37,17,20,40,18,38	12				
G9	9C	35	4,23,2,21,24,5,27,25,31,30,8,29,13, 34,32,11,15,37,17,20,40,18,38	36				
G10	10C	37	4,23,2,21,24,5,27,25,31,30,8,29,13, 34,32,11,35,15,17,20,40,18,38	16				
G11	11C	20	4,23,2,21,24,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,40,18,38	19				
G12	12C	18	4,23,2,21,24,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,38	39				
G13	1C	23	4,2,21,24,5,27,25,31,30,8,29,13,34, 32,11,35,15,37,17,20,40,18,38	3				
G14	2C	21	4,23,2,24,5,27,25,31,30,8,29,13,34, 32,11,35,15,37,17,20,40,18,38	22				

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TABLE IX. Output voltage (high-level, V_{DD} = 5V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G15	3C Output voltage, high level, V_{DD} = 5V	5	4,23,2,21,24,27,25,31,30,8,29,13,34, 32,11,35,15,37, 7,20,40,18,38	26	4.6	5.0	V	9
G16	4C	25	4,23,2,21,24,5,27,31,30,8,29,13,34, 32,11,35,15,37, 7,20,40,18,38	6				
G17	5C	30	4,23,2,21,24,5,27,25,31,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	9				
G18	6C	29	4,23,2,21,24,5,27,25,31,30,8,13,34, 32,11,35,15,37,7,20,40,18,38	28				
G19	7C	34	4,23,2,21,24,5,27,25,31,30,8,29, 13,32,11,35,15,37,17,20,40,18,38	33				
G20	8C	11	4,23,2,21,24,5,27,25,31,30,8,29,34, 13,32,35,15,37,17,20,40,18,38	12				
G21	9C	15	4,23,2,21,24,5,27,25,31,30,8,29,34, 13,32,11,35,37,17,20,40,18,38	36				
G22	10C	17	4,23,2,21,24,5,27,25,31,30,8,29,34, 13,32,11,35,15,37,20,,40,18,38	16				
G23	11C	40	4,23,2,21,24,5,27,25,31,30,8,29,34, 13,32,11,35,15,37,17,20,18,38	19				
G24	12C	38	4,23,2,21,24,5,27,25,31,30,8,29,34, 13,32,11,35,15,37,20,40	39				

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NOTES:
1. R1 = R2 = 51.1Ω ±5%, 1/2 W.
2. R3 = @ -55°C 18.2 kΩ ±1%, 1/8 W.
   @ 0° and 25°C 23.2 kΩ ±1%, 1/8 W.
   @ 60°C 27.4 kΩ ±1%, 1/8 W.
   @ 100°C 32 4 kΩ ±1%, 1/8 W.

FIGURE 9. Output voltage (high-level, V_{DD} = 5V) test circuit for table IX.

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TABLE X. Output voltage (low-level, V_{DD} = 15V) test.

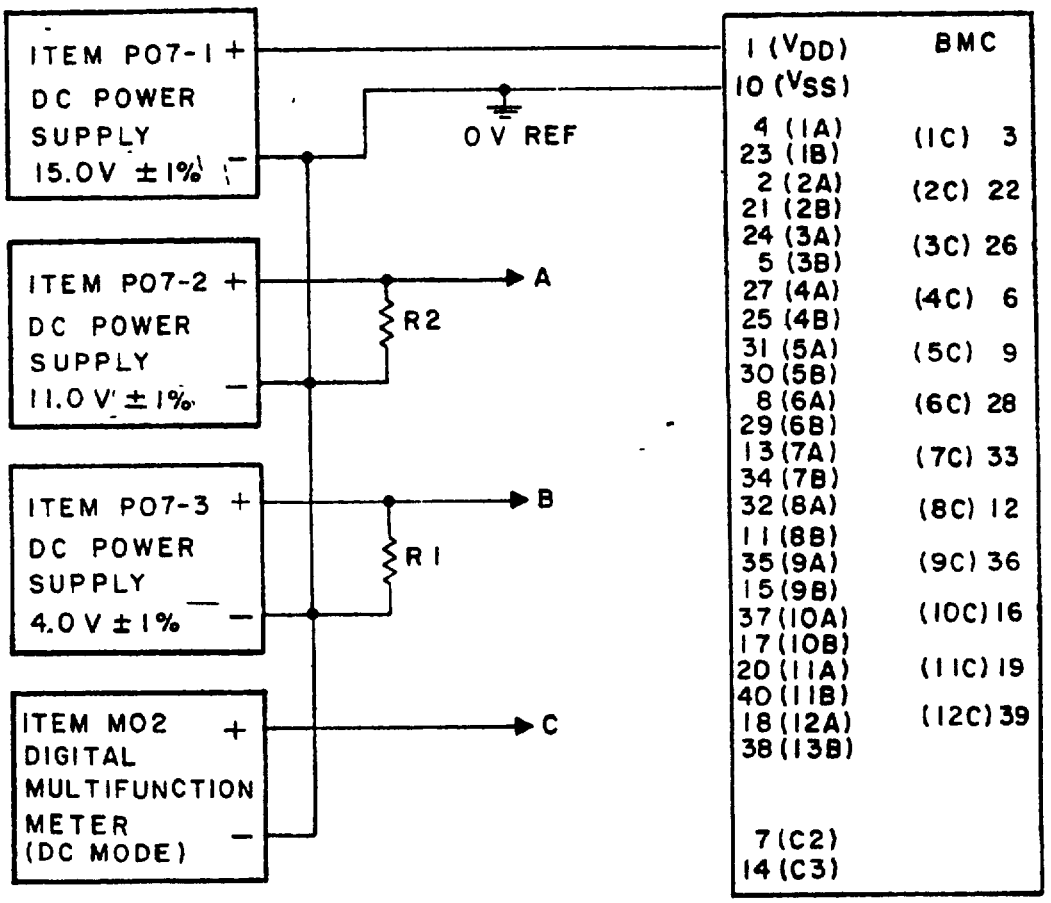
Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H1	1C Output voltage, low level, V_{DD} = 15V	Open	4,23,2,21,24,5,27,25, 31,30,8,29,13,34,32, 11,35,15,37,17,20,40 18,38	3	0.0	1.5	V	10
H2	2C			22				
H3	3C			26				
H4	4C			6				
H5	5C			9				
H6	6C			28				
H7	7C			33				
H8	8C			12				
H9	9C			36				
H10	10C			16				
H11	11C			19				
H12	12C			39				

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TABLE XI. Output voltage (high-level, V_{DD} = 15V) test.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
J1	1C Output voltage, high level, V_{DD} = 15V	4,2,24,27,31,8,13,32,35,37,20,18	23,21,5,25,30,29,34,11,15,17,40,38	3	13.5	15.0	V	10
J2	2C			22				
J3	3C			26				
J4	4C			6				
J5	5C			9				
J6	6C			28				
J7	7C			33				
J8	8C			12				
J9	9C			36				
J10	10C			16				
J11	11C			19				
J12	12C			39				

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NOTES:
1. R1 = 51.1Ω ±5%, 1/2 W.
2. R2 = 576Ω ±5%, 1/4 W.

FIGURE 10. Output voltage (low-level and high-level, V_{DD} = 15 V) test circuit for tables X and XI.

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TABLE XII. Input current (low-level) test.

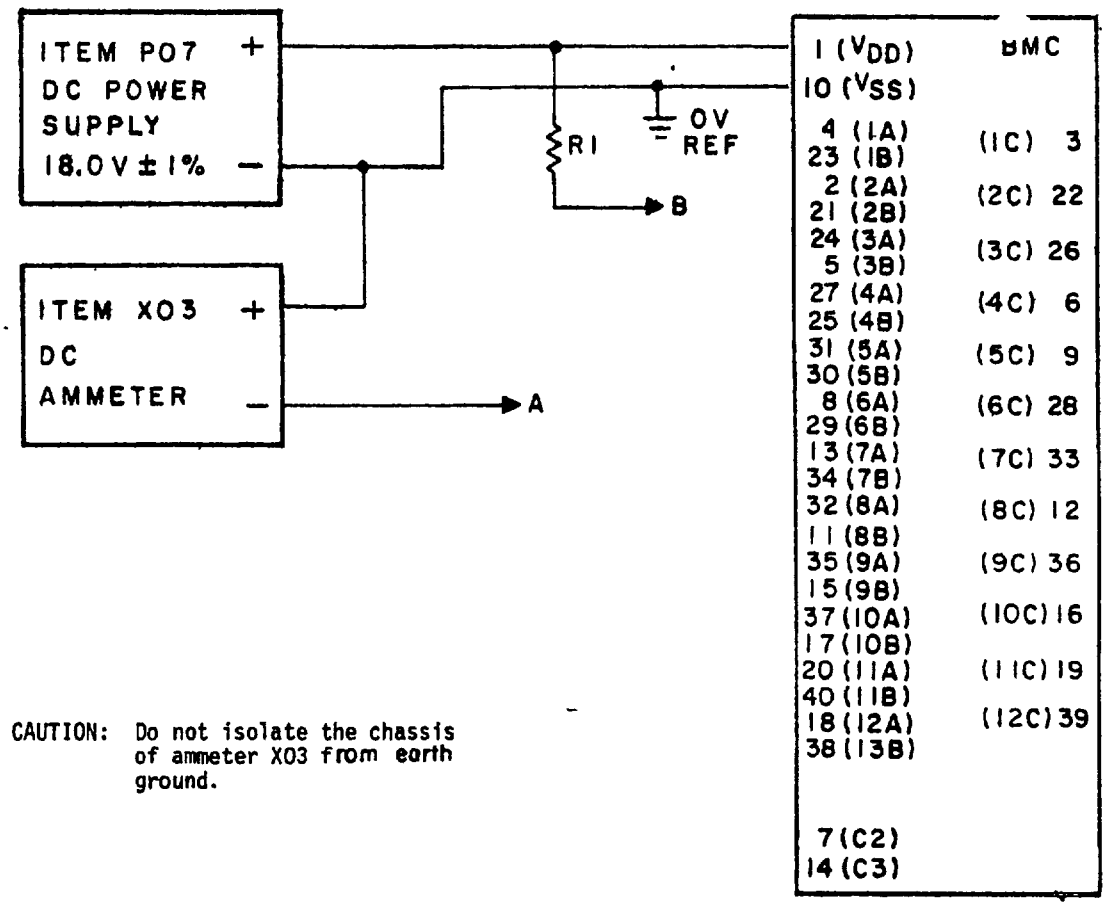
Test condition	Parameter	Connect: A to pin	Connect: B to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
K1	1A Input current, low level	4	23,2,21,24,5,27,25,31,30,8,29,13,34,32,11, 35,15,37,17,20 40,18,38	0	-200	nA	11
K2	1B	23	4,2,21,24,5,27,25,31,30,8,29,13,34,32,11, 35,15,37,17,20,40,18,38				
K3	2A	2	4,23,21,24,5,27,25,31,30,8,29,13,34,32,11, 11,35,15,37,17,20,40,18,38				
K4	2B	21	4,23,2,24,5,27,25,31,30,8,29,13,34,32,11, 35,15,37,17,20,40,18,38				
K5	3A	24	4,23,2,21,5,27,25,31,30,8,29,13,34,32,11, 35,15,37,17,20,40,18,38				
K6	3B	5	4,23,2,21,24,27,25,31,30,8,29,13,34,32,11, 35,15,37,17,20,40,18,38				
K7	4A	27	4,23,2,21,24,5,25,31,30,8,29,13,34,32,11, 35,15,37,17,20,40,18,38				
K8	4B	25	4,23,2,21,24,5,27,31,30,8,29,13,34,32,11, 35,15,37,17,20,40,18,38				
K9	5A	31	4,23,2,21,24,5,27,25,30,8,29,13,34,32,11, 35,15,37,17,20,40,18,38				
K10	5B	30	4,23,2,21,24,5,27,25,31,8,29,13,34,32,11, 35,15,37,17,20,40,18,38				
K11	6A	8	4,23,2,21,24,5,27,25,31,30,29,13,34,32,11, 35,15,37,17,20,40,18,38				
K12	6B	29	4,23,2,21,24,5,27,25,31,30,8,13,34,32,11, 35,15,37,17,20,40,18,38				
K13	7A	13	4,23,2,21,24,5,27,25,31,30,8,29,34,32,11, 35,15,37,17,20,40,18,38				
K14	7B	34	4,23,2,21,24,5,27,25,31,30,8,29,13,32,11, 35,15,37,17,20,40,18,38				

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TABLE XII. Input current (low-level) test - Continued.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
K15	8A Input current, low level	32	4,23,2,21,24,5,27,25,31,30,8,29,13,34,11, 35,15,37,17,20,40,18,38	0	-200	nA	11
K16	8B	11	4,23,2,21,24,5,27,25,31,30,8,29,13,34,32, 35,15,37,17,20,40,18,38				
K17	9A	35	4,23,2,21,24,5,27,25,31,30,8,29,13,34,32, 11,15,37,17,20,40,18,38				
K18	9B	15	4,23,2,21,24,5,27,25,31,30,8,29,13,34,32, 11,35,37,17,20,40,18,38				
K19	10A	37	4,23,2,21,24,5,27,25,31,30,8,29,13,34,32, 11,35,15,17,20,40,18,38				
K20	10B	17	4,23,2,21,24,5,27,25,31,30,8,29,13,34,32, 11,35,15,37,20,40,18,38				
K21	11A	20	4,23,2,21,24,5,27,25,31,30,8,29,13,34,32, 11,35,15,37,17,40,18,38				
K22	11B	40	4,23,2,21,24,5,27,25,31,30,8,29,13,34,32, 11,35,15,37,17,20,18,38				
K23	12A	18	4,23,2,21,24,5,27,25,31,30,8,29,13,34,32, 11,35,15,37,17,20,40,38				
K24	12B	38	4,23,2,21,24,5,27,25,31,30,8,29,13,34,32, 11,35,15,37,17,20,40,18				

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CAUTION: Do not isolate the chassis of ammeter X03 from earth ground.

NOTE: R1 = 1.5 kΩ ±1%, 1/4 W.

FIGURE 11. Input current (low-level) test circuit for table XII

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TABLE XIII. Input current (high-level) test.

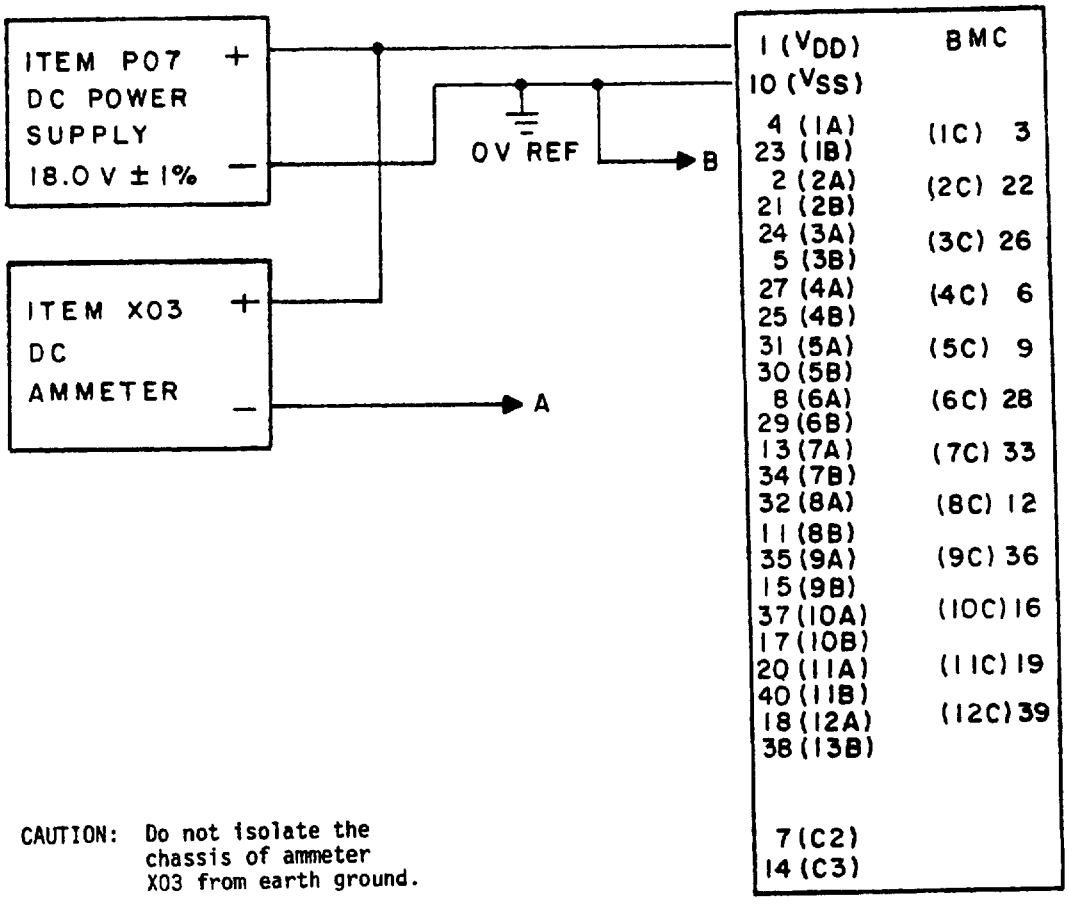
Test condition		Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
L1	1A	Input current, high level	4	23,2,21,24,5,27,25,31,30,8,29,13,34,32,11, 35,15,37,17,20 40,18,38	0	200	nA	12
L2	1B		23	4,2,21,24,5,27,25,31,30,8,29,13,34,32,11, 35,15,37,17,20,40,18,38				
L3	2A		2	4,23,21,24,5,27,25,31,30,8,29,13,34,32, 11,35,15,37,17,20,40,18,38				
L4	2B		21	4,23,2,24,5,27,25,31,30,8,29,13,34,32,11, 35,15,37,17,20,40,18,38				
L5	3A		24	4,23,2,21,5,27,25,31,30,8,29,13,34,32,11, 35,15,37,17,20,40,18,38				
L6	3B		5	4,23,2,21,24,27,25,31,30,8,29,13,34,32,11, 35,15,37,17,20,40,18,38				
L7	4A		27	4,23,2,21,24,5,25,31,30,8,29,13,34,32,11, 35,15,37,17,20,40,18,38				
L8	4B		25	4,23,2,21,24,5,27,31,30,8,29,13,34,32,11, 35,15,37,17,20,40,18,38				
L9	5A		31	4,23,2,21,24,5,27,25,30,8,29,13,34,32,11, 35,15,37,17,20,40,18,38				
L10	5B		30	4,23,2,21,24,5,27,25,31,8,29,13,34,32,11, 35,15,37,17,20,40,18,38				
L11	6A		8	4,23,2,21,24,5,27,25,31,30,29,13,34,32,11, 35,15,37,17,20,40,18,38				
L12	6B		29	4,23,2,21,24,5,27,25,31,30,8,13,34,32,11, 35,15,37,17,20,40,18,38				
L13	7A		13	4,23,2,21,24,5,27,25,31,30,8,29,34,32,11, 35,15,37,17,20,40,18,38				
L14	7B		34	4,23,2,21,24,5,27,25,31,30,8,29,13,32,11, 35,15,37,17,20,40,18,38				

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TABLE XIII. Input current (high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
L15	8A Input current, high level	32	4,23,2,21,24,5,27,25,31,30,8,29,13,34,11, 35,15,37,17,20,40,18,38	0	200	nA	12
L16	8B	11	4,23,2,21,24,5,27,25,31,30,8,29,13,34,32, 35,15,37,17,20,40,18,38				
L17	9A	35	4,23,2,21,24,5,27,25,31,30,8,29,13,34,32, 11,15,37,17,20,40,18,38				
L18	9B	15	4,23,2,21,24,5,27,25,31,30,8,29,13,34,32, 11,35,37,17,20,40,18,38				
L19	10A	37	4,23,2,21,24,5,27,25,31,30,8,29,13,34,32, 11,35,15,17,20,40,18,38				
L20	10B	17	4,23,2,21,24,5,27,25,31,30,8,29,13,34,32, 11,35,15,37,20,40,18,38				
L21	11A	20	4,23,2,21,24,5,27,25,31,30,8,29,13,34,32, 11,35,15,37,17,40,18,38				
L22	11B	40	4,23,2,21,24,5,27,25,31,30,8,29,13,34,32, 11,35,15,37,17,20,18,38				
L23	12A	18	4,23,2,21,24,5,27,25,31,30,8,29,13,34,32, 11,35,15,37,17,20,40,38				
L24	12B	38	4,23,2,21,24,5,27,25,31,30,8,29,13,34,32, 11,35,15,37,17,20,40,18				

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CAUTION: Do not isolate the chassis of ammeter X03 from earth ground.

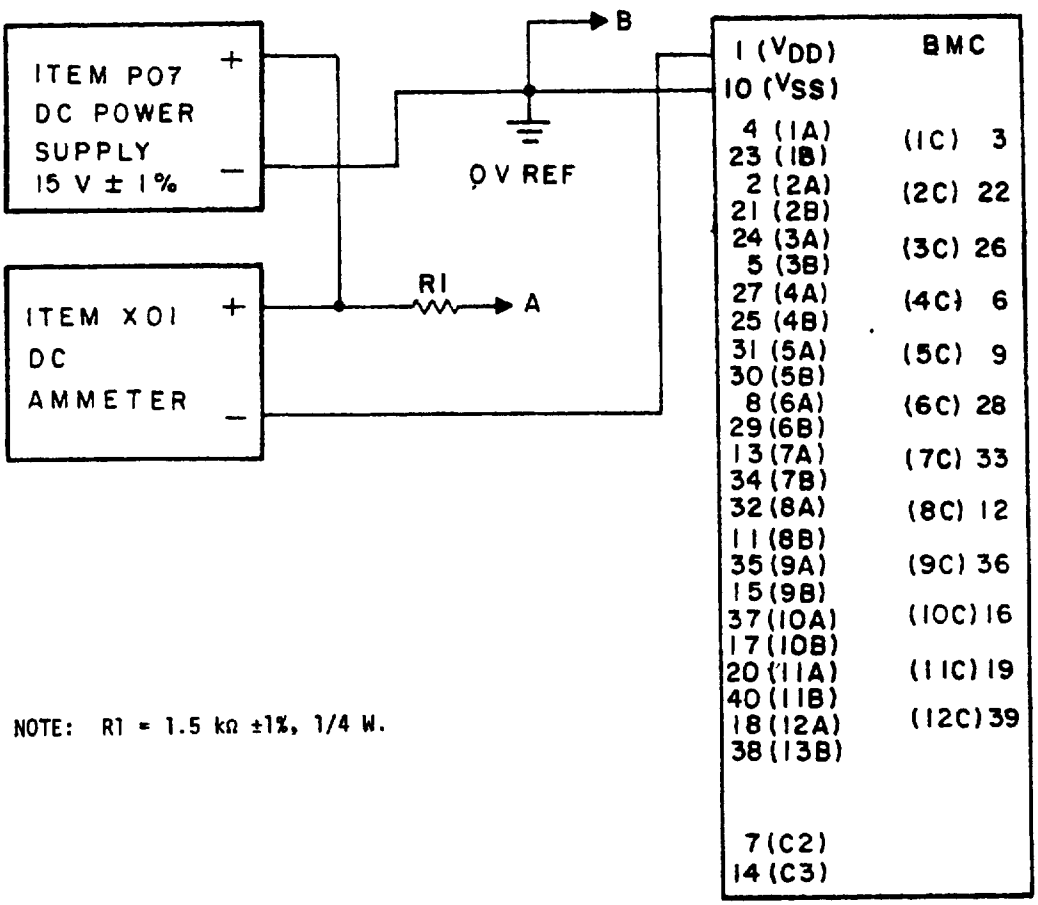
FIGURE 12 Input current (high-level) test circuit for table XIII.

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TABLE XIV. Power supply current (low-level and high-level) test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M1	Power supply current, low level	Open	4,23,2,21,24, 5,27,25,31,30, 8,29,13,34,32, 11,35,15,37,17, 20,40,18,38	0	6	µA	13
M2	Power supply current, high level	4,2,24,27,31,8, 13,32,35,37,20, 18	23,21,5,25,30, 29,34,11,15,17, 40,38	0	6	µA	13

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NOTE: R1 = 1.5 kΩ ±1%, 1/4 W.

CAUTION: Do not isolate the chassis of ammeter X01 from earth ground.

FIGURE 13. Power supply current (low-level and high-level) test circuit for table XIV.

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TABLE XV. Transition time (t_{TLH}, V_{DD} = 5V) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
N1	1C Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 5V	23,2,21,24,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	4	3	20	200	ns	14,15
N2	2C	4,23,21,24,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	2	22				
N3	3C	4,23,2,21,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	24	26				
N4	4C	4,23,2,21,24,5,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	27	6				
N5	5C	4,23,2,21,24,5,27,25,30,8,29, 13,34,32,11,35,15,37,17,20,40,18,38	31	9				
N6	6C	4,23,2,21,24,5,27,25,31,30,29,13, 34,32,11,35,15,37,17,20,40,18,38	8	28				
N7	7C	4,23,2,21,24,5,27,25,31,30,8,29, 34,32,11,35,15,37,17,20,40,18,38	13	33				
N8	8C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,11,35,15,37,17,20,40,18,38	32	12				
N9	9C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,15,37,17,20,40,18,38	35	36				
N10	10C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,17,20,40,18,38	37	16				
N11	11C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,17,40,18,38	20	19				
N12	12C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,17,20,40,38	18	39				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:06Z
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TABLE XV. Transition time (t_{TLH}, V_{DD} = 5V) test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
N13	1C Transition time (t_{TLH}), C_L = 150 pF, V_{DD} = 5V	23,2,21,24,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	4	3	68	680	ns	14,15
N14	2C	4,23,21,24,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	2	22				
N15	3C	4,23,2,21,5,27,25,31,30,8,29,13,34, 32,11,35,15,37,17,20,40,18,38	24	26				
N16	4C	4,23,2,21,24,5,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	27	6				
N17	5C	4,23,2,21,24,5,27,25,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	31	9				
N18	6C	4,23,2,21,24,5,27,25,31,30,29,13, 34,32,11,35,15,37,17,20,40,18,38	8	28				
N19	7C	4,23,2,21,24,5,27,25,31,30,8,29, 34,32,11,35,15,37,17,20,40,18,38	13	33				
N20	8C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,11,35,15,37,17,20,40,18,38	32	12				
N21	9C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,15,37,17,20,40,18,38	35	36				
N22	10C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,17,20,40,18,38	37	16				
N23	11C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,17,40,18,38	20	19				
N24	12C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,17,20,40,38	18	39				

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TABLE XVI. Transition time (t_{THL}, V_{DD} = 5V) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
P1	1C Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 5V	23,2,21,24,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	4	3	20	200	ns	14,15
P2	2C	4,23,21,24,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	2	22				
P3	3C	4,23,2,21,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	24	26				
P4	4C	4,23,2,21,24,5,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	27	6				
P5	5C	4,23,2,21,24,5,27,25,30,8,29, 13,34,32,11,35,15,37,17,20,40,18,38	31	9				
P6	6C	4,23,2,21,24,5,27,25,31,30,29,13, 34,32,11,35,15,37,17,20,40,18,38	8	28				
P7	7C	4,23,2,21,24,5,27,25,31,30,8,29, 34,32,11,35,15,37,17,20,40,18,38	13	33				
P8	8C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,11,35,15,37,17,20,40,18,38	32	12				
P9	9C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,15,37,17,20,40,18,38	35	36				
P10	10C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,17,20,40,18,38	37	16				
P11	11C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,17,40,18,38	20	19				
P12	12C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,17,20,40,38	18	39				

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TABLE XVI. Transition time (t_{THL}, V_{DD} = 5V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
P13	1C Transition time (t_{THL}), C_L = 150 pF, V_{DD} = 5V	23,2,21,24,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	4	3	68	680	ns	14,15
P14	2C	4,23,21,24,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	2	22				
P15	3C	4,23,2,21,5,27,25,31,30,8,29,13,34, 32,11,35,15,37,17,20,40,18,38	24	26				
P16	4C	4,23,2,21,24,5,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	27	6				
P17	5C	4,23,2,21,24,5,27,25,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	31	9				
P18	6C	4,23,2,21,24,5,27,25,31,30,29,13, 34,32,11,35,15,37,17,20,40,18,38	8	28				
P19	7C	4,23,2,21,24,5,27,25,31,30,8,29, 34,32,11,35,15,37,17,20,40,18,38	13	33				
P20	8C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,11,35,15,37,17,20,40,18,38	32	12				
P21	9C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,15,37,17,20,40,18,38	35	36				
P22	10C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,17,20,40,18,38	37	16				
P23	11C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,17,40,18,38	20	19				
P24	12C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,17,20,40,38	18	39				

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TABLE XVII. Transition time (t_{TLH}, V_{DD} = 10V) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Q1	1C Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 10V	23,2,21,24,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	4	3	10	100	ns	14,15
Q2	2C	4,23,21,24,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	2	22				
Q3	3C	4,23,2,21,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	24	26				
Q4	4C	4,23,2,21,24,5,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	27	6				
Q5	5C	4,23,2,21,24,5,27,25,30,8,29, 13,34,32,11,35,15,37,17,20,40,18,38	31	9				
Q6	6C	4,23,2,21,24,5,27,25,31,30,29,13, 34,32,11,35,15,37,17,20,40,18,38	8	28				
Q7	7C	4,23,2,21,24,5,27,25,31,30,8,29, 34,32,11,35,15,37,17,20,40,18,38	13	33				
Q8	8C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,11,35,15,37,17,20,40,18,38	32	12				
Q9	9C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,15,37,17,20,40,18,38	35	36				
Q10	10C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,17,20,40,18,38	37	16				
Q11	11C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,17,40,18,38	20	19				
Q12	12C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,17,20,40,38	18	39				

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TABLE XVIII. Transition time (t_{THL}, V_{DD} = 10V) test.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
R1	1C Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 10V	23,2,21,24,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	4	3	10	100	ns	14,15
R2	2C	4,23,21,24,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	2	22				
R3	3C	4,23,2,21,5,27,25,31,30,8,29,13,34, 32,11,35,15,37,17,20,40,18,38	24	26				
R4	4C	4,23,2,21,24,5,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	27	6				
R5	5C	4,23,2,21,24,5,27,25,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	31	9				
R6	6C	4,23,2,21,24,5,27,25,31,30,29,13, 34,32,11,35,15,37,17,20,40,18,38	8	28				
R7	7C	4,23,2,21,24,5,27,25,31,30,8,29, 34,32,11,35,15,37,17,20,40,18,38	13	33				
R8	8C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,11,35,15,37,17,20,40,18,38	32	12				
R9	9C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,15,37,17,20,40,18,38	35	36				
R10	10C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,17,20,40,18,38	37	16				
R11	11C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,17,40,18,38	20	19				
R12	12C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,17,20,40,38	18	39				

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TABLE XIX. Transition time (t_{TLH}, V_{DD} = 15V) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
S1	1C Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 15V	23,2,21,24,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	4	3	8	80	ns	14,15
S2	2C	4,23,21,24,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	2	22				
S3	3C	4,23,2,21,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	24	26				
S4	4C	4,23,2,21,24,5,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	27	6				
S5	5C	4,23,2,21,24,5,27,25,30,8,29, 13,34,32,11,35,15,37,17,20,40,18,38	31	9				
S6	6C	4,23,2,21,24,5,27,25,31,30,29,13, 34,32,11,35,15,37,17,20,40,18,38	8	28				
S7	7C	4,23,2,21,24,5,27,25,31,30,8,29, 34,32,11,35,15,37,17,20,40,18,38	13	33				
S8	8C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,11,35,15,37,17,20,40,18,38	32	12				
S9	9C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,15,37,17,20,40,18,38	35	36				
S10	10C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,17,20,40,18,38	37	16				
S11	11C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,17,40,18,38	20	19				
S12	12C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,17,20,40,38	18	39				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:06Z
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TABLE XX. Transition time (t_{THL}, V_{DD} = 15V) test.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
T1	1C Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 15V	23,2,21,24,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	4	3	8	80	ns	14,15
T2	2C	4,23,21,24,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	2	22				
T3	3C	4,23,2,21,5,27,25,31,30,8,29,13,34, 32,11,35,15,37,17,20,40,18,38	24	26				
T4	4C	4,23,2,21,24,5,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	27	6				
T5	5C	4,23,2,21,24,5,27,25,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	31	9				
T6	6C	4,23,2,21,24,5,27,25,31,30,29,13, 34,32,11,35,15,37,17,20,40,18,38	8	28				
T7	7C	4,23,2,21,24,5,27,25,31,30,8,29, 34,32,11,35,15,37,17,20,40,18,38	13	33				
T8	8C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,11,35,15,37,17,20,40,18,38	32	12				
T9	9C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,15,37,17,20,40,18,38	35	36				
T10	10C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,17,20,40,18,38	37	16				
T11	11C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,17,40,18,38	20	19				
T12	12C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,17,20,40,38	18	39				

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TABLE XXI. Propagation delay time (t_{PLH}, V_{DD} = 5V) test.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
U1	1A to 1C Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5V	23,2,21,24,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	4	3	35	350	ns	14,15
U2	1B to 1C	4,2,21,24,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	23	3				
U3	2A to 2C	4,23,21,24,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	2	22				
U4	2B to 2C	4,23,2,24,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	21	22				
U5	3A to 3C	4,23,2,21,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	24	26				
U6	3B to 3C	4,23,2,21,24,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	5	26				
U7	4A to 4C	4,23,2,21,24,5,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	27	6				
U8	4B to 4C	4,23,2,21,24,5,27,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	25	6				
U9	5A to 5C	4,23,2,21,24,5,27,25,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	31	9				
U10	5B to 5C	4,23,2,21,24,5,27,25,31,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	30	9				
U11	6A to 6C	4,23,2,21,24,5,27,25,31,30,29,13, 34,32,11,35,15,37,17,20,40,18,38	8	28				
U12	6B to 6C	4,23,2,21,24,5,27,25,31,30,8,13, 34,32,11,35,15,37,17,20,40,18,38	29	28				

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TABLE XXI. Propagation delay time (t_{PLH}, V_{DD} = 5V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
U13	7A to 7C Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5V	4,23,2,21,24,5,27,25,31,30,8,29, 34,32,11,35,15,37,17,20,40,18,38	13	33	35	350	ns	14,15
U14	7B to 7C	4,23,2,21,24,5,27,25,31,30,8,29, 13,32,11,35,15,37,17,20,40,18,38	34	33				
U15	8A to 8C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,11,35,15,37,17,20,40,18,38	32	12				
U16	8B to 8C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,35,15,37,17,20,40,18,38	11	12				
U17	9A to 9C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,15,37,17,20,40,18,38	35	36				
U18	9B to 9C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,37,17,20,40,18,38	15	36				
U19	10A to 10C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,17,20,40,18,38	37	16				
U20	10B to 10C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,20,40,18,38	17	16				
U21	11A to 11C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,17,40,18,38	20	19				
U22	11B to 11C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,17,20,18,38	40	19				
U23	12A to 12C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,17,20,40,38	18	39				
U24	12B to 12C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,17,20,40,18	38	39				

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TABLE XXI. Propagation delay time (t_{PLH}, V_{DD} = 5V) test - Continued.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
U25	1A to 1C Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5V	23,2,21,24,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	4	3	55	550	ns	14,15
U26	1B to 1C	4,2,21,24,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	23	3				
U27	2A to 2C	4,23,21,24,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	2	22				
U28	2B to 2C	4,23,2,24,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	21	22				
U29	3A to 3C	4,23,2,21,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	24	26				
U30	3B to 3C	4,23,2,21,24,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	5	26				
U31	4A to 4C	4,23,2,21,24,5,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	27	6				
U32	4B to 4C	4,23,2,21,24,5,27,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	25	6				
U33	5A to 5C	4,23,2,21,24,5,27,25,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	31	9				
U34	5B to 5C	4,23,2,21,24,5,27,25,31,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	30	9				
U35	6A to 6C	4,23,2,21,24,5,27,25,31,30,29,13, 34,32,11,35,15,37,17,20,40,18,38	8	28				
U36	6B to 6C	4,23,2,21,24,5,27,25,31,30,8,13, 34,32,11,35,15,37,17,20,40,18,38	29	28				

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TABLE XXI. Propagation delay time (t_{PLH}, V_{DD} = 5V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
U37	7A to 7C Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5V	4,23,2,21,24,5,27,25,31,30,8,29, 34,32,11,35,15,37,17,20,40,18,38	13	33	55	550	ns	14,15
U38	7B to 7C	4,23,2,21,24,5,27,25,31,30,8,29, 13,32,11,35,15,37,17,20,40,18,38	34	33				
U39	8A to 8C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,11,35,15,37,17,20,40,18,38	32	12				
U40	8B to 8C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,35,15,37,17,20,40,18,38	11	12				
U41	9A to 9C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,15,37,17,20,40,18,38	35	36				
U42	9B to 9C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,37,17,20,40,18,38	15	36				
U43	10A to 10C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,17,20,40,18,38	37	16				
U44	10B to 10C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,20,40,18,38	17	16				
U45	11A to 11C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,17,40,18,38	20	19				
U46	11B to 11C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,17,20,18,38	40	19				
U47	12A to 12C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,17,20,40,38	18	39				
U48	12B to 12C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,17,20,40,18	38	39				

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TABLE XXII. Propagation delay time (t_{PHL}, V_{DD} = 5V) test.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
V1	1A to 1C Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5V	23,2,21,24,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	4	3	35	350	ns	14,15
V2	1B to 1C	4,2,21,24,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	23	3				
V3	2A to 2C	4,23,21,24,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	2	22				
V4	2B to 2C	4,23,2,24,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	21	22				
V5	3A to 3C	4,23,2,21,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	24	26				
V6	3B to 3C	4,23,2,21,24,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	5	26				
V7	4A to 4C	4,23,2,21,24,5,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	27	6				
V8	4B to 4C	4,23,2,21,24,5,27,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	25	6				
V9	5A to 5C	4,23,2,21,24,5,27,25,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	31	9				
V10	5B to 5C	4,23,2,21,24,5,27,25,31,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	30	9				
V11	6A to 6C	4,23,2,21,24,5,27,25,31,30,29,13, 34,32,11,35,15,37,17,20,40,18,38	8	28				
V12	6B to 6C	4,23,2,21,24,5,27,25,31,30,8,13, 34,32,11,35,15,37,17,20,40,18,38	29	28				

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TABLE XXII. Propagation delay time (t_{PHL}, V_{DD} = 5V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
V13	7A to 7C Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5V	4,23,2,21,24,5,27,25,31,30,8,29, 34,32,11,35,15,37,17,20,40,18,38	13	33	35	350	ns	14,15
V14	7B to 7C	4,23,2,21,24,5,27,25,31,30,8,29, 13,32,11,35,15,37,17,20,40,18,38	34	33				
V15	8A to 8C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,11,35,15,37,17,20,40,18,38	32	12				
V16	8B to 8C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,35,15,37,17,20,40,18,38	11	12				
V17	9A to 9C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,15,37,17,20,40,18,38	35	36				
V18	9B to 9C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,37,17,20,40,18,38	15	36				
V19	10A to 10C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,17,20,40,18,38	37	16				
V20	10B to 10C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,20,40,18,38	17	16				
V21	11A to 11C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,17,40,18,38	20	19				
V22	11B to 11C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,17,20,18,38	40	19				
V23	12A to 12C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,17,20,40,38	18	39				
V24	12B to 12C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,17,20,40,18	38	39				

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TABLE XXII. Propagation delay time (t_{PHL}, V_{DD} = 5V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
V25	1A to 1C Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5V	3,2,21,24,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	4	3	55	550	ns	14,15
V26	1B to 1C	4,2,21,24,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	23	3				
V27	2A to 2C	4,23,21,24,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	2	22				
V28	2B to 2C	4,23,2,24,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	21	22				
V29	3A to 3C	4,23,2,21,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	24	26				
V30	3B to 3C	4,23,2,21,24,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	5	26				
V31	4A to 4C	4,23,2,21,24,5,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	27	6				
V32	4B to 4C	4,23,2,21,24,5,27,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	25	6				
V33	5A to 5C	4,23,2,21,24,5,27,25,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	31	9				
V34	5B to 5C	4,23,2,21,24,5,27,25,31,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	30	9				
V35	6A to 6C	4,23,2,21,24,5,27,25,31,30,29,13, 34,32,11,35,15,37,17,20,40,18,38	8	28				
V36	6B to 6C	4,23,2,21,24,5,27,25,31,30,8,13, 34,32,11,35,15,37,17,20,40,18,38	29	28				

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TABLE XXII. Propagation delay time (t_{PHL}, V_{DD} = 5V) test - Continued.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
V37	7A to 7C Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5V	4,23,2,21,24,5,27,25,31,30,8,29, 34,32,11,35,15,37,17,20,40,18,38	13	33	55	550	ns	14,15
V38	7B to 7C	4,23,2,21,24,5,27,25,31,30,8,29, 13,32,11,35,15,37,17,20,40,18,38	34	33				
V39	8A to 8C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,11,35,15,37,17,20,40,18,38	32	12				
V40	8B to 8C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,35,15,37,17,20,40,18,38	11	12				
V41	9A to 9C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,15,37,17,20,40,18,38	35	36				
V42	9B to 9C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,37,17,20,40,18,38	15	36				
V43	10A to 10C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,17,20,40,18,38	37	16				
V44	10B to 10C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,20,40,18,38	17	16				
V45	11A to 11C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,17,40,18,38	20	19				
V46	11B to 11C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,17,20,18,38	40	19				
V47	12A to 12C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,17,20,40,38	18	39				
V48	12B to 12C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,17,20,40,18	38	39				

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TABLE XXIII. Propagation delay time (t_{PLH}, V_{DD} = 10V) test.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
W1	1A to 1C Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10V	23,2,21,24,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	4	3	15	150	ns	14,15
W2	1B to 1C	4,2,21,24,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	23	3				
W3	2A to 2C	4,23,21,24,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	2	22				
W4	2B to 2C	4,23,2,24,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	21	22				
W5	3A to 3C	4,23,2,21,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	24	26				
W6	3B to 3C	4,23,2,21,24,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	5	26				
W7	4A to 4C	4,23,2,21,24,5,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	27	6				
W8	4B to 4C	4,23,2,21,24,5,27,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	25	6				
W9	5A to 5C	4,23,2,21,24,5,27,25,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	31	9				
W10	5B to 5C	4,23,2,21,24,5,27,25,31,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	30	9				
W11	6A to 6C	4,23,2,21,24,5,27,25,31,30,29,13, 34,32,11,35,15,37,17,20,40,18,38	8	28				
W12	6B to 6C	4,23,2,21,24,5,27,25,31,30,8,13, 34,32,11,35,15,37,17,20,40,18,38	29	28				

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TABLE XXIII Propagation delay time (t_{PLH}, V_{DD} = 10V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
W13	7A to 7C Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10V	4,23,2,21,24,5,27,25,31,30,8,29, 34,32,11,35,15,37,17,20,40,18,38	13	33	15	150	ns	14,15
W14	7B to 7C	4,23,2,21,24,5,27,25,31,30,8,29, 13,32,11,35,15,37,17,20,40,18,38	34	33				
W15	8A to 8C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,11,35,15,37,17,20,40,18,38	32	12				
W16	8B to 8C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,35,15,37,17,20,40,18,38	11	12				
W17	9A to 9C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,15,37,17,20,40,18,38	35	36				
W18	9B to 9C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,37,17,20,40,18,38	15	36				
W19	10A to 10C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,17,20,40,18,38	37	16				
W20	10B to 10C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,20,40,18,38	17	16				
W21	11A to 11C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,17,40,18,38	20	19				
W22	11B to 11C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,17,20,18,38	40	19				
W23	12A to 12C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,17,20,40,38	18	39				
W24	12B to 12C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,17,20,40,18	38	39				

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TABLE XXIV. Propagation delay time (t_{PHL}, V_{DD} = 10V) test.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
X1	1A to 1C Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10V	23,2,21,24,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	4	3	15	150	ns	14,15
X2	1B to 1C	4,2,21,24,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	23	3				
X3	2A to 2C	4,23,21,24,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	2	22				
X4	2B to 2C	4,23,2,24,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	21	22				
X5	3A to 3C	4,23,2,21,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	24	26				
X6	3B to 3C	4,23,2,21,24,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	5	26				
X7	4A to 4C	4,23,2,21,24,5,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	27	6				
X8	4B to 4C	4,23,2,21,24,5,27,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	25	6				
X9	5A to 5C	4,23,2,21,24,5,27,25,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	31	9				
X10	5B to 5C	4,23,2,21,24,5,27,25,31,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	30	9				
X11	6A to 6C	4,23,2,21,24,5,27,25,31,30,29,13, 34,32,11,35,15,37,17,20,40,18,38	8	28				
X12	6B to 6C	4,23,2,21,24,5,27,25,31,30,8,13, 34,32,11,35,15,37,17,20,40,18,38	29	28				

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TABLE XXIV. Propagation delay time (t_{PHL}, V_{DD} = 10V) test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
X13	7A to 7C Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10V	4,23,2,21,24,5,27,25,31,30,8,29, 34,32,11,35,15,37,17,20,40,18,38	13	33	15	150	ns	14,15
X14	7B to 7C	4,23,2,21,24,5,27,25,31,30,8,29, 13,32,11,35,15,37,17,20,40,18,38	34	33				
X15	8A to 8C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,11,35,15,37,17,20,40,18,38	32	12				
X16	8B to 8C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,35,15,37,17,20,40,18,38	11	12				
X17	9A to 9C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,15,37,17,20,40,18,38	35	36				
X18	9B to 9C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,37,17,20,40,18,38	15	36				
X19	10A to 10C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,17,20,40,18,38	37	16				
X20	10B to 10C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,20,40,18,38	17	16				
X21	11A to 11C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,17,40,18,38	20	19				
X22	11B to 11C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,17,20,18,38	40	19				
X23	12A to 12C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,17,20,40,38	18	39				
X24	12B to 12C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,17,20,40,18	38	39				

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TABLE XXV. Propagation delay time (t_{PLH}, V_{DD} = 15V) test.

Test condition	Parameter		Connect: A to pin	Connect: B to pin	Connect: C to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
Y1	1A to 1C	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15V	23,2,21,24,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	4	3	10	100	ns	14,15
Y2	1B to 1C		4,2,21,24,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	23	3				
Y3	2A to 2C		4,23,21,24,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	2	22				
Y4	2B to 2C		4,23,2,24,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	21	22				
Y5	3A to 3C		4,23,2,21,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	24	26				
Y6	3B to 3C		4,23,2,21,24,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	5	26				
Y7	4A to 4C		4,23,2,21,24,5,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	27	6				
Y8	4B to 4C		4,23,2,21,24,5,27,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	25	6				
Y9	5A to 5C		4,23,2,21,24,5,27,25,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	31	9				
Y10	5B to 5C		4,23,2,21,24,5,27,25,31,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	30	9				
Y11	6A to 6C		4,23,2,21,24,5,27,25,31,30,29,13, 34,32,11,35,15,37,17,20,40,18,38	8	28				
Y12	6B to 6C		4,23,2,21,24,5,27,25,31,30,8,13, 34,32,11,35,15,37,17,20,40,18,38	29	28				

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TABLE YXIV. Propagation delay time (t_{PLH}, V_{DD} = 15V) test - Continued.

Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
Y13	7A to 7C Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15V	4,23,2,21,24,5,27,25,31,30,8,29, 34,32,11,35,15,37,17,20,40,18,38	13	33	10	100	ns	14,15
Y14	7B to 7C	4,23,2,21,24,5,27,25,31,30,8,29, 13,32,11,35,15,37,17,20,40,18,38	34	33				
Y15	8A to 8C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,11,35,15,37,17,20,40,18,38	32	12				
Y16	8B to 8C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,35,15,37,17,20,40,18,38	11	12				
Y17	9A to 9C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,15,37,17,20,40,18,38	35	36				
Y18	9B to 9C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,37,17,20,40,18,38	15	36				
Y19	10A to 10C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,17,20,40,18,38	37	16				
Y20	10B to 10C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,20,40,18,38	17	16				
Y21	11A to 11C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,17,40,18,38	20	19				
Y22	11B to 11C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,17,20,18,38	40	19				
Y23	12A to 12C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,17,20,40,38	18	39				
Y24	12B to 12C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,17,20,40,18	38	39				

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TABLE XXVI. Propagation delay time (t_{PHL}, V_{DD} = 15V) test.

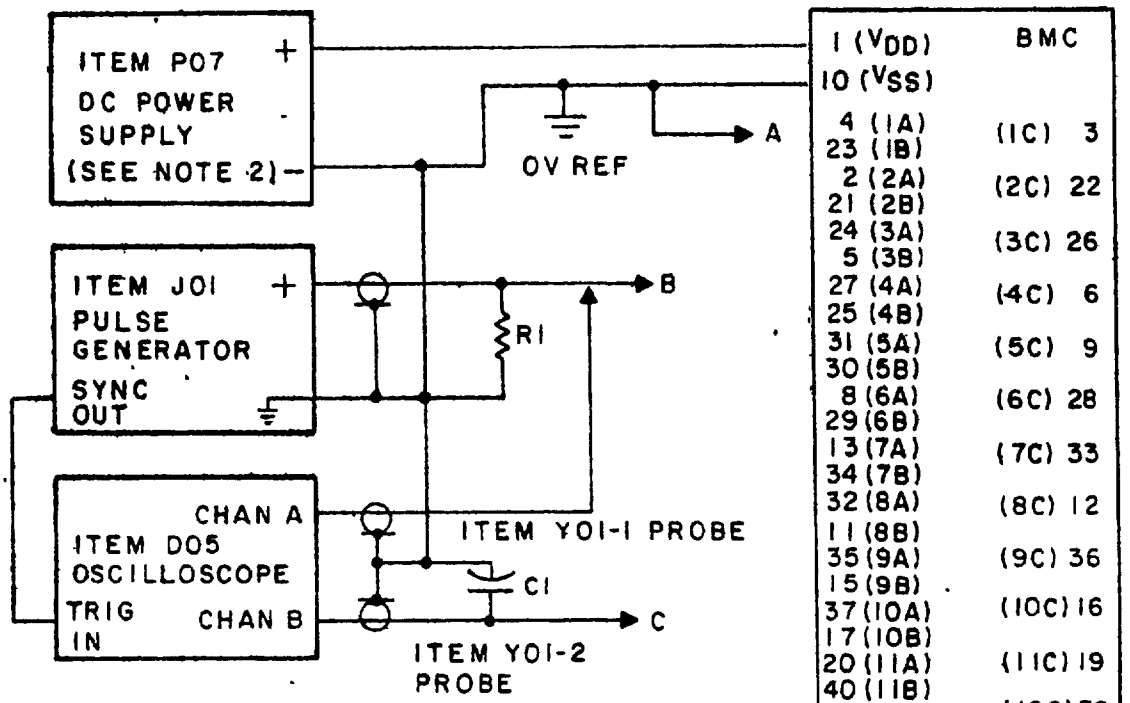
Test condition	Parameter	Connect: A to pin	Connect: B to pin	Connect: C to pin	Initial limits 25°±5°C ambient: Min	Initial limits 25°±5°C ambient: Max	Unit	Test figure
Z1	1A to 1C Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15V	23,2,21,24,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	4	3	10	100	ns	14,15
Z2	1B to 1C	4,2,21,24,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	23	3				
Z3	2A to 2C	4,23,21,24,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	2	22				
Z4	2B to 2C	4,23,2,24,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	21	22				
Z5	3A to 3C	4,23,2,21,5,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	24	26				
Z6	3B to 3C	4,23,2,21,24,27,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	5	26				
Z7	4A to 4C	4,23,2,21,24,5,25,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	27	6				
Z8	4B to 4C	4,23,2,21,24,5,27,31,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	25	6				
Z9	5A to 5C	4,23,2,21,24,5,27,25,30,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	31	9				
Z10	5B to 5C	4,23,2,21,24,5,27,25,31,8,29,13, 34,32,11,35,15,37,17,20,40,18,38	30	9				
Z11	6A to 6C	4,23,2,21,24,5,27,25,31,30,29,13, 34,32,11,35,15,37,17,20,40,18,38	8	28				
Z12	6B to 6C	4,23,2,21,24,5,27,25,31,30,8,13, 34,32,11,35,15,37,17,20,40,18,38	29	28				

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TABLE XXVI. Propagation delay time (t_{PHL}, V_{DD} = 15V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Z13	7A to 7C Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15V	4,23,2,21,24,5,27,25,31,30,8,29, 34,32,11,35,15,37,17,20,40,18,38	13	33	10	100	ns	14,15
Z14	7B to 7C	4,23,2,21,24,5,27,25,31,30,8,29, 13,32,11,35,15,37,17,20,40,18,38	34	33				
Z15	8A to 8C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,11,35,15,37,17,20,40,18,38	32	12				
Z16	8B to 8C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,35,15,37,17,20,40,18,38	11	12				
Z17	9A to 9C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,15,37,17,20,40,18,38	35	36				
Z18	9B to 9C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,37,17,20,40,18,38	15	36				
Z19	10A to 10C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,17,20,40,18,38	37	16				
Z20	10B to 10C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,20,40,18,38	17	16				
Z21	11A to 11C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,17,40,18,38	20	19				
Z22	11B to 11C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,17,20,18,38	40	19				
Z23	12A to 12C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,17,20,40,38	18	39				
Z24	12B to 12C	4,23,2,21,24,5,27,25,31,30,8,29, 13,34,32,11,35,15,37,17,20,40,18	38	39				

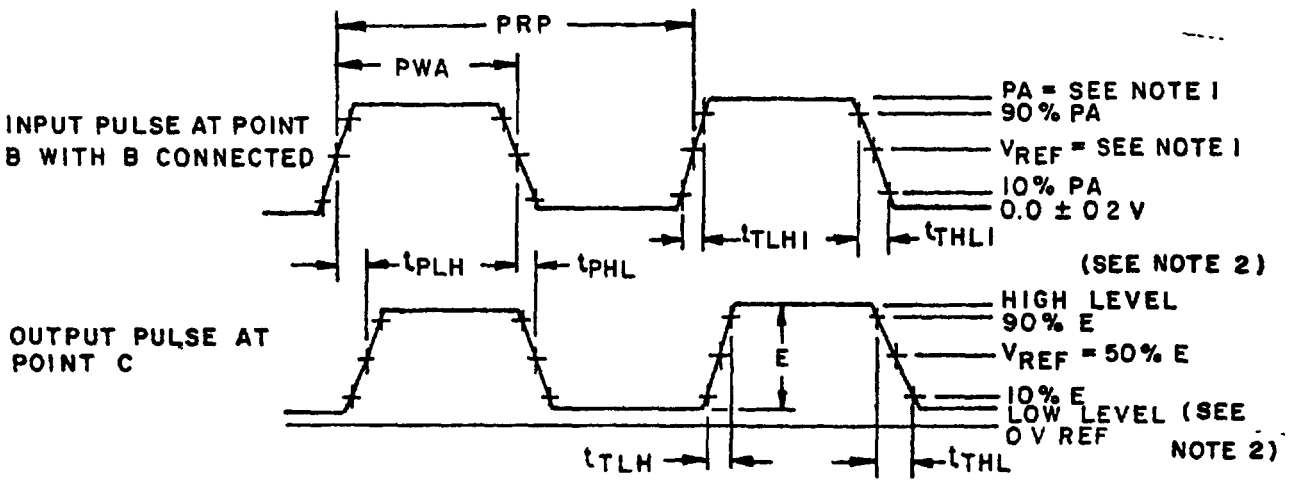
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NOTES:
1. R1 = 51.1Ω ±5%, 1/2 W when V_{DD} = 5 V.
   2 W when V_{DD} = 10 V.
   5 W when V_{DD} = 15 V.
2. Adjust item P07 voltage as follows:
   V_{DD} 5 V ±1% for test conditions N1-N24, P1-P24, U1-U48 and V1-V48.
   V_{DD} = 10 V ±1% for test conditions Q1-Q12, R1-R12, W1-W24 and X1-X24.
   V_{DD} = 15 V ±1% for test conditions S1-S12, T1-T12, Y1-Y24 and Z1-Z24.
3. C1 = 50 ±5 pF including probe and parasitic capacitance for test conditions N1-N12, P1-P12, Q1-Q12, R1-R12, S1-S12, T1-T12, U1-U24, V1-V24, W1-W24, X1-X24, Y1-Y24 and Z1-Z24.
   = 150 ±15 pF including probe and parasitic capacitance for test conditions N13-N24, P13-P24, U25-U48 and V25-V48.

FIGURE 14. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XV through XXVI

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NOTES:

1. Input pulse parameters:
   - PA = 5.0 ±0.1 V for test conditions N1-N24, P1-P24, U1-U48 and V1-V48.
     = 10.0 ±0.1 V for test conditions Q1-Q12, R1-R12, W1-W24 and X1-X24.
     = 15.0 ±0.1 V for test conditions S1-S12, T1-T12, Y1-Y24 and Z1-Z24.
   - V_{REF} = 2.5 ±0.1 V for test conditions N1-N24, P1-P24, U1-U48 and V1-V48.
     = 5.0 ±0.1 V for test conditions Q1-Q12, R1-R12, W1-W24 and X1-X24.
     = 7.5 ±0.1 V for test conditions S1-S12, T1-T12, Y1-Y24 and Z1-Z24.
   - PWA = 2.5 ±0.2 µs for test conditions N1-N24, P1-P24, U1-U48 and V1-V48.
     = 1.0 ±0.1 µs for test conditions Q1-Q12, R1-R12, S1-S12, T1-T12, W1-W24, X1-X24, Y1-Y24 and Z1-Z24.
   - PRP = 5.0 ±0.5 µs.
   - $t_{TLH1} = t_{THL1}$ = 10 ±2 ns.
2. Before making the measurements verify that the output pulse high level and low level are as follows:

V_{DD} =	High level	low level
5.0 V	≥ 4.6 V	≤ 0.4 V
10.0 V	≥ 9.5 V	≤ 0.5 V
15.0 V	≥ 13.5 V	≤ 1.5 V

3. The lead length of electrical conductors between resistor R1 and input pins, probes Y01-1 and Y01-2 to input and output pins, and between load circuit and output pin shall be less than one inch (25.4 mm).
4. All voltages are referenced to V_{SS}.

FIGURE 15 Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XV through XXVI.

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TABLE XXVII. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
AA	Power supply filter capacitance	Open	1	0.007	0.013	µF	16

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TABLE XXVIII. C2 and C3 capacitance test.

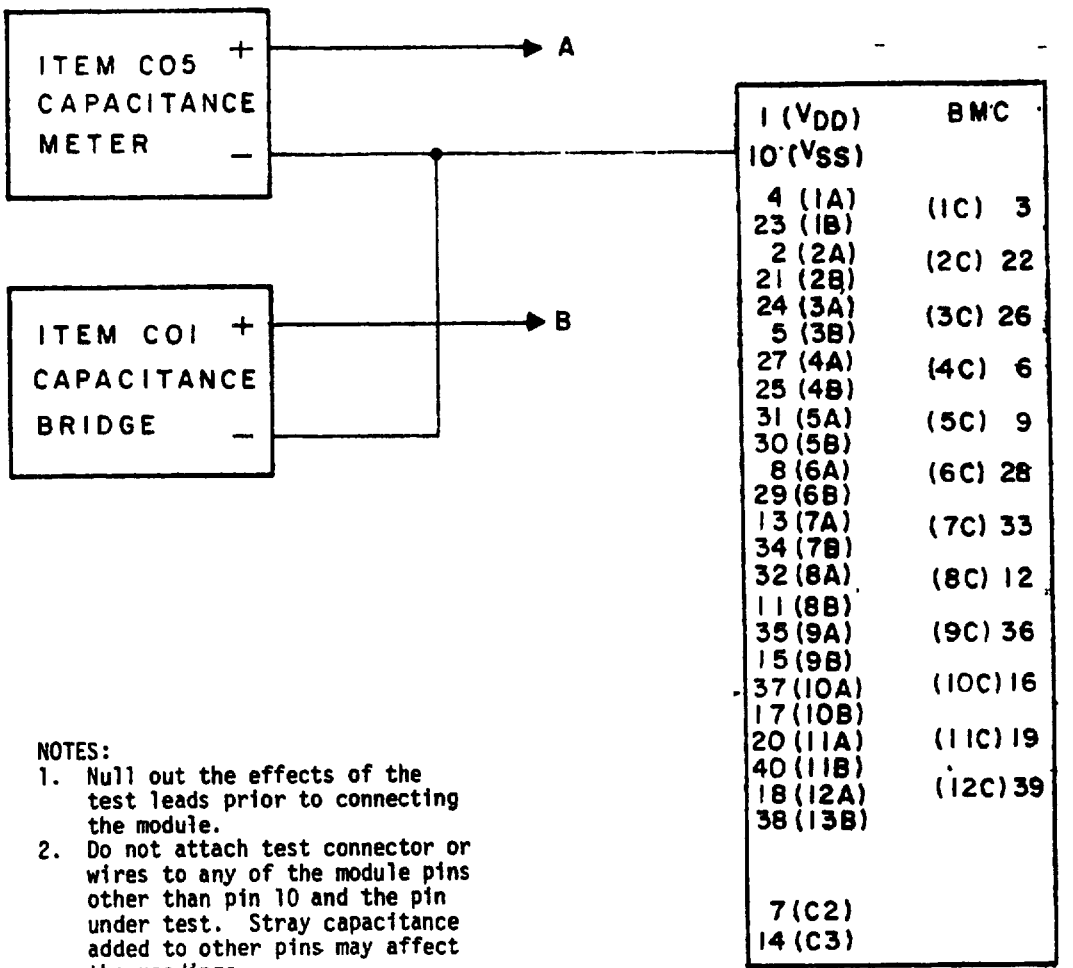
Test condition	Parameter	Connect		Initial limits 25°±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
BB1	C2 Capacitance	7	Open	176	264	pF	16
BB2	C3 Capacitance	14	Open	176	264	pF	16

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TABLE XXIX. Input capacitance test.

Test condition	Parameter	Connect A to pin	Connect B to pin	Initial limits 25°±5°C ambient Min	Initial limits 25°±5°C ambient Max	Unit	Test figure
CC1	1A Input capacitance	4	Open	2	15	pF	16
CC2	1B	23					
CC3	2A	2					
CC4	2B	21					
CC5	3A	24					
CC6	3B	5					
CC7	4A	27					
CC8	4B	25					
CC9	5A	31					
CC10	5B	30					
CC11	6A	8					
CC12	6B	29					
CC13	7A	13					
CC14	7B	34					
CC15	8A	32					
CC16	8B	11					
CC17	9A	35					
CC18	9B	15					
CC19	10A	37					
CC20	10B	17					
CC21	11A	20					
CC22	11B	40					
CC23	12A	18					
CC24	12B	38					

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NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Capacitance bridge (CO1) setup:

   Frequency = 1 kHz.
   V_{pk} = 0.3 V maximum.

FIGURE 16. Power supply filter, C2 and C3, and input capacitance test circuit for tables XXVII, XXVIII and XXIX.

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